

Rwanda's Industrial Shift

A Target Strategy for Complexity,
Capability, and Growth





Executive Summary

The report Rwanda's Industrial Shift: A Target Strategy for Complexity, Capability, and Growth provides a comprehensive, evidence-based framework for advancing Rwanda's structural transformation through export diversification and capability development. Drawing on the Economic Complexity Index Optimization (ECI-OPT) model, it identifies a portfolio of 52 products that can realistically expand Rwanda's productive base while maximizing long-term gains in economic sophistication.

Rwanda's current growth model, while successful in delivering strong aggregate expansion, remains structurally narrow and concentrated in a limited set of primary exports, notably minerals, coffee, and tea. This concentration exposes the economy to commodity price volatility and constrains the depth of domestic value addition. Despite progress in export growth, the country's Economic Complexity Index (ECI) remains relatively low at -0.45 , reflecting limited diversification and a modest accumulation of productive capabilities.

The report emphasizes that economic complexity is a critical determinant of long-term development outcomes. Economies that diversify into more sophisticated products tend to experience faster and more growth, improved resilience to external shocks, and better social outcomes, including lower inequality and emissions intensity. Accordingly, the report advocates for a strategic approach to diversification based on "relatedness," whereby new industries are developed by building on existing capabilities, ensuring that each step reinforces the next in a cumulative process of industrial upgrading.

In order to translate this framework into actionable priorities, the report identifies eight flagship product areas across chemicals, plastics, electronics, and medical devices. These are organized into a sequenced pathway comprising foundational capability upgrades, adjacent growth opportunities, and strategic innovation bets. This structure allows Rwanda to begin with feasible, lower-risk activities such as blending, packaging, and quality control and progressively move toward more complex and knowledge-intensive production systems, including precision manufacturing and regulated industries.

The analysis further presents the importance of grounding diversification in Rwanda's structural realities. As a landlocked economy, Rwanda must prioritize products with sufficiently high value-to-weight ratios to remain competitive despite elevated transport costs. At the same time, capability constraints require focusing on products that are within reach of existing skills, infrastructure, and institutional capacity. These feasibility thresholds ensure that the diversification strategy remains both pragmatic and scalable.

If effectively implemented, the proposed strategy could significantly strengthen Rwanda's economic trajectory. The report projects that successful entry into the identified product portfolio could raise the country's ECI to positive territory, accelerate GDP per capita growth, reduce emissions intensity, and enhance resilience to external shocks. More broadly, the transition toward a capability-driven growth model would support the creation of higher-quality jobs, deepen industrial linkages, and expand the country's fiscal and innovation space.

Finally, the report argues that Rwanda's long-term prosperity will depend not only on the pace of growth but on its quality and composition. Moving from a commodity and volume-driven model toward one anchored in knowledge, capabilities, and industrial sophistication is essential for achieving sustained, inclusive, and resilient development. This transformation will require coordinated policy action, targeted investments, and strong institutional support to ensure that diversification translates into lasting structural change.

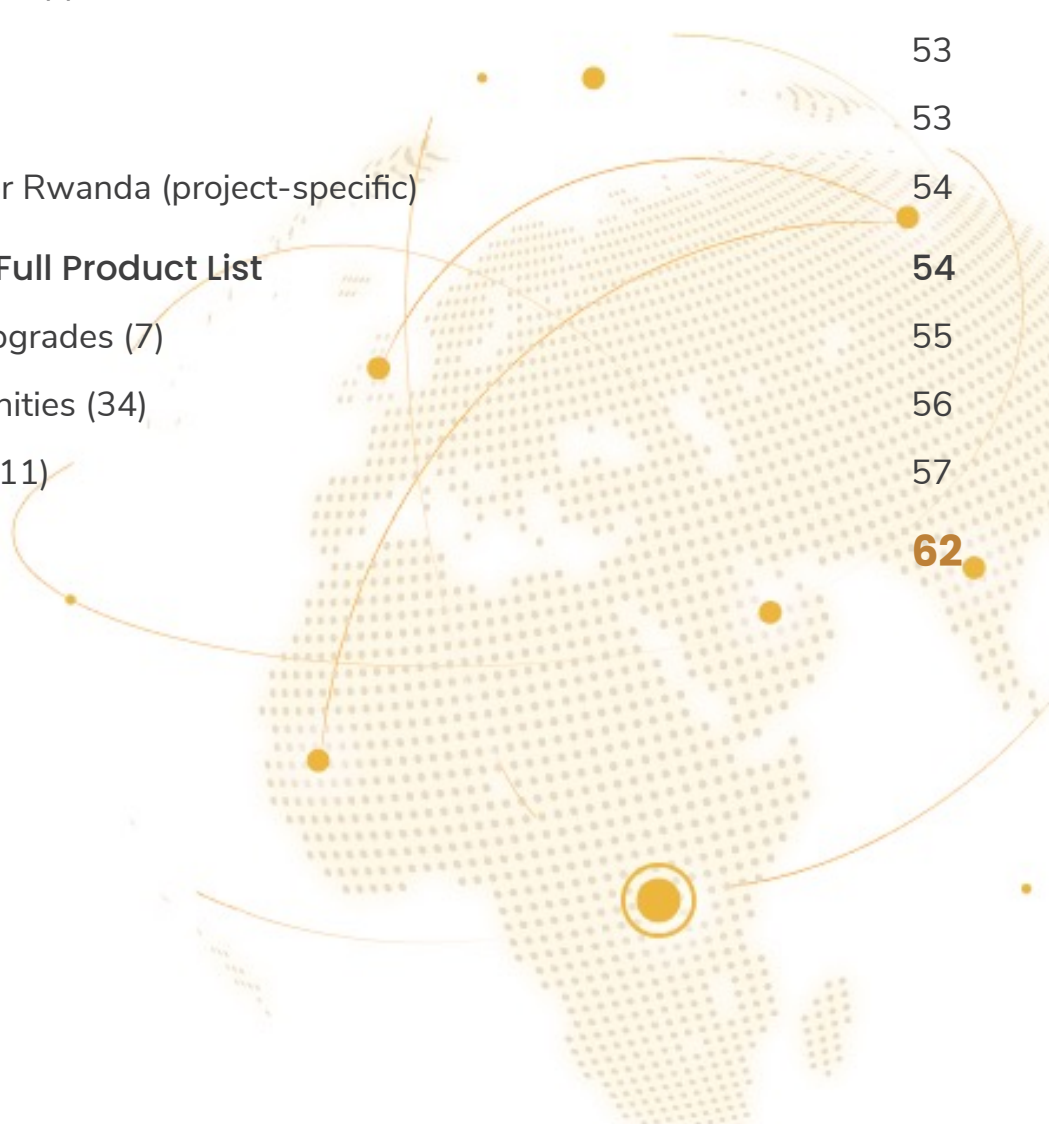
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List of Acronyms and Abbreviations

Acronym	Full Form
AfCFTA	African Continental Free Trade Area
AI	Artificial Intelligence
AIMS	African Institute for Mathematical Sciences
ASTM	American Society for Testing and Materials
BACI	Base pour l'Analyse du Commerce International (CEPII trade database)
BNR	National Bank of Rwanda (Banque Nationale du Rwanda)
BRD	Development Bank of Rwanda (Banque Rwandaise de Développement)
CAGR	Compound Annual Growth Rate
CMU	Carnegie Mellon University
EAC	East African Community
ECI	Economic Complexity Index
ECI-OPT	Economic Complexity Index Optimization Model
EU	European Union
FDA	Food and Drug Administration (United States)
GDP	Gross Domestic Product
HS	Harmonized System (international product classification)
IMTS	International Merchandise Trade Statistics
ISO	International Organization for Standardization
LFS	Labour Force Survey
MINECOFIN	Ministry of Finance and Economic Planning (Rwanda)
MIT	Massachusetts Institute of Technology

MS	Master of Science
NISR	National Institute of Statistics of Rwanda
OEC	Observatory of Economic Complexity
OEM	Original Equipment Manufacturer
PCI	Product Complexity Index
PPP	Purchasing Power Parity
PSSF	Private Sector Support Fund
QA	Quality Assurance
QC	Quality Control
RCA	Revealed Comparative Advantage
RDB	Rwanda Development Board
RDGF	Rwanda Development Guiding Fund
RSB	Rwanda Standards Board
SADC	Southern African Development Community
SEZ	Special Economic Zone
SITC	Standard International Trade Classification
UNDP	United Nations Development Programme
US / USD	United States / United States Dollar
WDI	World Development Indicators (World Bank)
WHO	World Health Organization
WHO-GMP	WHO Good Manufacturing Practice

Glossary of Key Terms

Term	Definition
Economic Complexity Index (ECI)	A score that summarises how sophisticated a country's exports are. It is higher when a country exports many products competitively, particularly products that few other countries can produce. Higher ECI is associated with faster long-run income growth, lower inequality and lower emissions per dollar of output. Rwanda's ECI was -0.45 (2023).
Product Complexity Index (PCI)	The product-level counterpart of the ECI. It scores each traded product based on the diversity and sophistication of the countries that export it competitively. High-PCI products (for example, medical instruments) are mainly exported by countries with broad, knowledge-intensive economies. Comparing a product's PCI with Rwanda's ECI provides a rough indication of the capability stretch required to enter that product.
Revealed Comparative Advantage (RCA)	A measure of export specialization. It compares a product's share in a country's exports with that product's share in world exports. If $RCA > 1$, the country exports that product more intensively than the world average, indicating relative specialization (often used as a practical threshold for being a competitive exporter).
Relatedness	A data-driven measure of how feasible it is for a country to become competitive in a new product, based on how similar that product is to what it already exports. It is inferred from global co-export patterns in the product space and interpreted as shared capabilities (inputs, know-how, skills and supporting institutions). Higher relatedness to Rwanda's current export basket usually indicates easier entry.
Product space	A map of products based on how often they are exported together across countries. Products that are close to one another tend to require similar capabilities, making countries more likely to diversify into nearby products. In this report, the product space is used to assess relatedness and identify feasible "next steps" for diversification.
Diversification frontier	A chart that plots potential new products by feasibility (how close they are to a country's current capabilities, often proxied by relatedness) and potential gain (how complex the product is, proxied by the PCI). It illustrates the trade-off between easier "nearby" entries and more demanding, higher-upgrading moves, and helps sequence diversification by prioritising products that balance entry likelihood with capability gains.
Capability	The practical know-how and systems needed to produce and export a product competitively. This includes skills, input, machinery, supplier networks, management routines, standards and certification, and the institutions that support them (for example, testing laboratories and regulators). Capabilities build over time and can be reused to enter related, more complex products.

Opportunity landscape	A structured view of the set of products a country could plausibly diversify into. It combines feasibility (how close products are to current capabilities, via relatedness) with potential impact (expected complexity gains and market potential) to prioritize and sequence diversification options.
ECI-OPT model	Developed by Stojkoski and Hidalgo (2025), a model that selects a portfolio of new products that can raise a country's ECI the most while remaining feasible given current capabilities. Feasibility is assessed using product space proximity (relatedness) and the estimated effort required to reach export competitiveness, typically defined as achieving $RCA \geq 1$. The output is a ranked, least-cost set of products that supports step-by-step upgrading.
Harmonized System (HS) codes	An international product classification used to label traded goods in customs and trade statistics. HS codes are standardized across countries and are usually reported at 2, 4 or 6 digits for increasing detail (with some countries using additional digits nationally). In this report, 4-digit HS codes are used to define products consistently in trade, RCA and complexity calculations.



Introduction

Economic Potential and Diversification Opportunities



Economic Potential and Diversification Opportunities

Rwanda's long-term development depends on raising productivity in existing sectors and expanding into more sophisticated activities. Global evidence shows that economies that export more complex products grow faster, sustain higher income levels and adapt more effectively to external shocks. This pattern is captured by the Economic Complexity Index (ECI) ^{1,3,5,6}, a measure of productive capabilities that is a strong predictor of long-term economic growth^{1,3,5,7–14}.

This report applies the Economic Complexity Optimisation (ECI-OPT) model (Stojkoski & Hidalgo, 2025) to identify these opportunities. The model produces a balanced portfolio of 52 products that maximises potential complexity gains while remaining feasible given Rwanda's current capabilities. It replaces sector heuristics (broad, pre-defined sector priority lists based on expert judgement rather than product-level evidence) with empirical optimisation, and highlights activities where incremental steps yield the strongest capability upgrades.

The resulting portfolio follows a coherent capability pathway. It begins with routine-intensive activities such as blending, packaging and quality control, which strengthen standardisation and process discipline. From there, it moves into plastics conversion, milling and chemical processing, creating entry points into precision manufacturing in regulated sectors.

To keep the diversification strategy grounded in Rwanda's operational realities, the portfolio recommended by the ECI-OPT model includes only products that meet basic feasibility criteria linked to logistics and capabilities. An explanation of these thresholds is provided in Chapter 1, where they are applied systematically to the opportunity set.

Overall, the analysis indicates clear scope for higher-value and more complex production. Advancing this product portfolio through coordinated policy action, blended finance and targeted skills development can support a shift towards capability-driven growth that raises job quality and strengthens competitiveness.

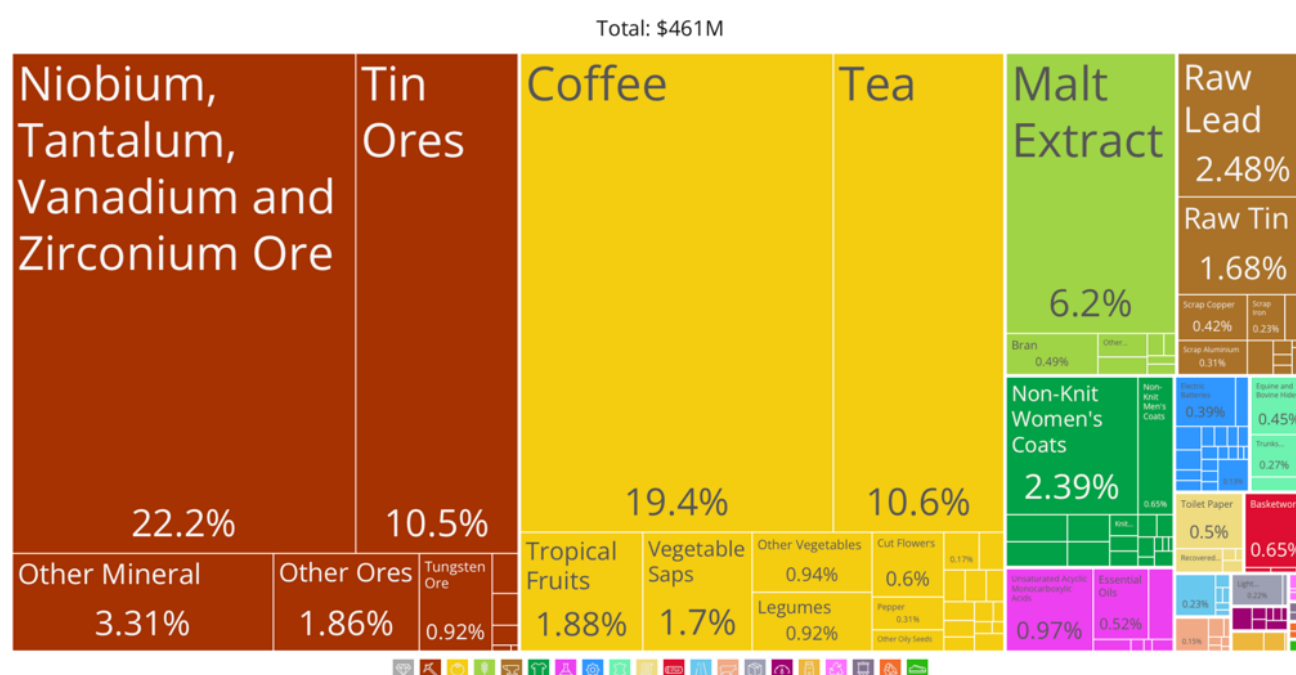
Chapter 1

The Economic Complexity of Rwanda

According to Rwanda's official external trade statistics, goods exports increased from US\$703 million in 2013 to US\$2.466 billion in 2023. Excluding gold, exports totaled US\$1.582 billion, highlighting the extent to which re-exported gold shapes the aggregate. These figures are drawn from the National Bank of Rwanda's Formal External Trade statistics, which are based on customs declarations compiled under IMTS 2010 and published mainly in SITC commodity sections.

Measuring economic complexity requires complete and comparable data. For this reason, the report uses the BACI-based Observatory of Economic Complexity (OEC) dataset, which provides a harmonised HS4 structure across trading partners, as well as the product-level detail needed to position Rwanda within the international product space. The analysis focuses on exports excluding gold to better reflect underlying domestic capabilities. On this basis, core non-gold exports reached US\$461 million in 2023, concentrated in coffee, tea, wolfram and coltan. These products reflect Rwanda's most established capabilities and form the platform from which broader diversification can occur.

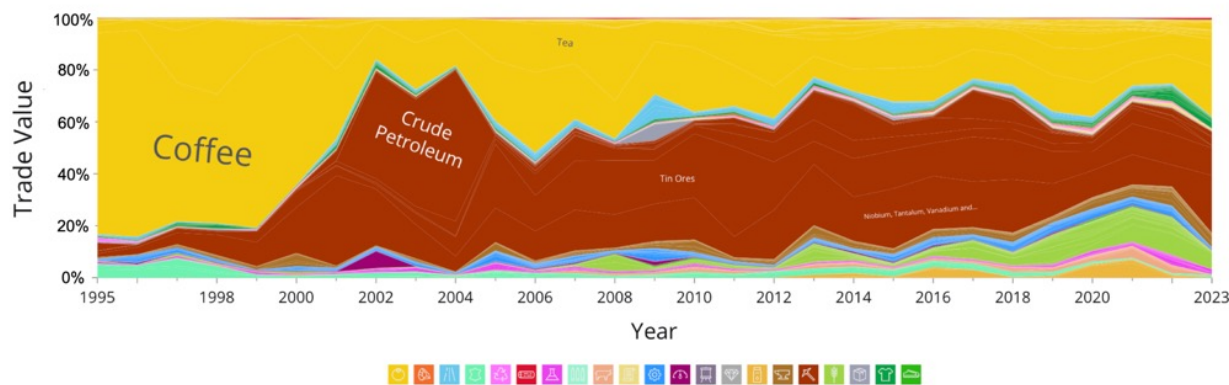
Figure 1. Core No-Gold Exports (BACI 2023): A Base for Diversification



Excluding gold, Rwanda's US\$461 million export basket is led by minerals, coffee and tea, with early footholds in food preparation, apparel, chemicals and light manufacturing. Together, these activities provide a foundation for further diversification.

Limited export diversification is consistent with Rwanda's current stage of industrial development. Over the past three decades, the country's export structure has shifted across a narrow set of primary products—coffee in the 1990s, refined petroleum re-exports in the 2000s and tin ores in the 2010s. Each surge contributed to short-term growth but left the production base concentrated in activities with modest spillovers and limited technological depth. More recently, small footholds have emerged in food preparation, apparel, chemicals and light manufacturing; however, these remain marginal relative to agriculture and minerals.

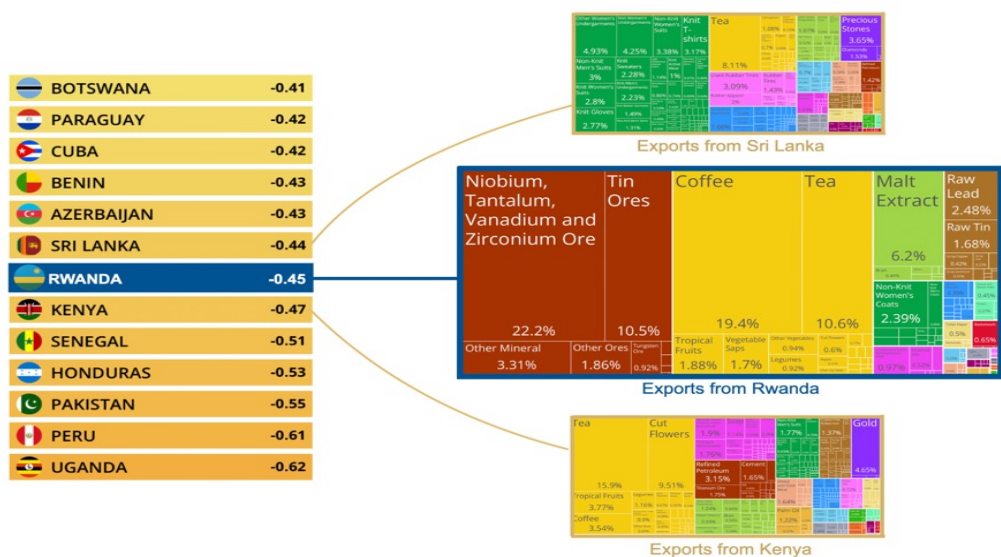
Figure 2. Core No-Gold Exports (1995–2023)



Manufactured exports have begun to emerge but remain small relative to minerals and agricultural products. Petroleum-related entries in the data correspond to re-exports of HS 2710; Rwanda does not export crude oil (HS 2709).

Despite these emerging signs of diversification, Rwanda’s economic complexity remains moderate. On a non-gold basis, the Economic Complexity Index (ECI) stood at –0.45 in 2023, a level typically associated with economies concentrated in a limited set of capabilities. This reflects an export structure in which a small number of primary products account for most export earnings, and where linkages to more knowledge-intensive goods remain limited.

Figure 3. ECI Rankings by Country (2023)



Higher positions reflect broader and more knowledge-intensive export bases among peer economies.

The result is a growth model that remains exposed to price volatility and is difficult to sustain. When commodity prices rise, activity accelerates; when they fall, vulnerability increases. Without structural change, Rwanda risks remaining dependent on a narrow productive base that generates foreign exchange but yields limited spillovers.

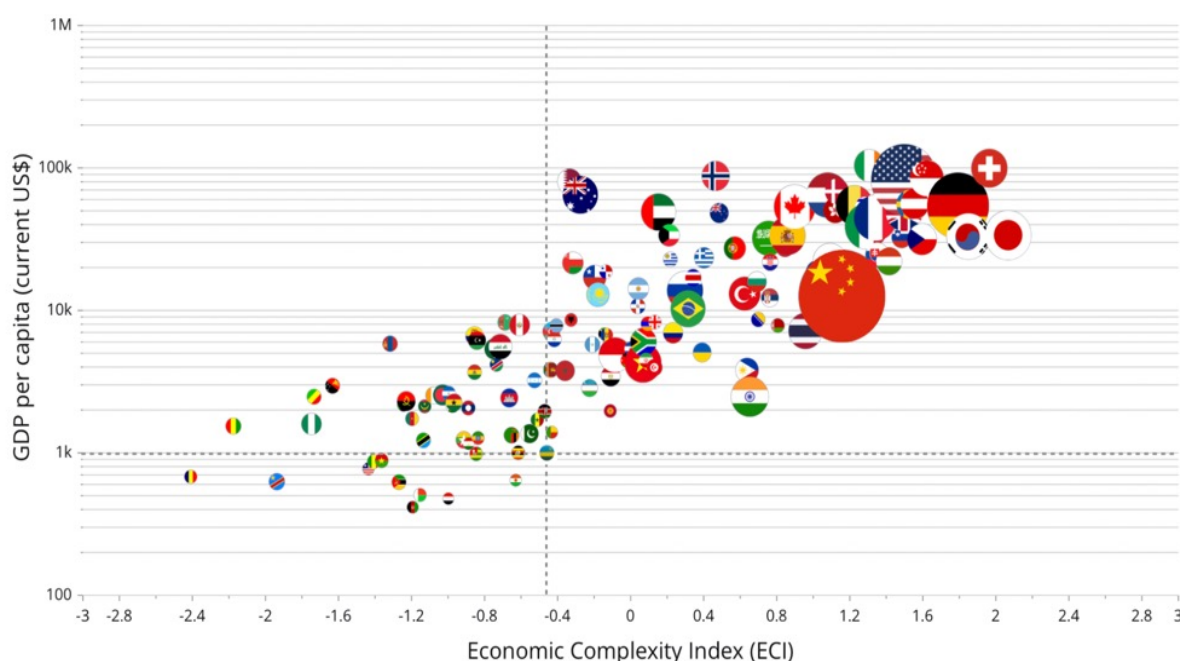
Why Complexity Matters

Economic complexity measures the knowledge embedded in a country's productive structure. The Economic Complexity Index (ECI) synthesizes how sophisticated a country's exports are, using trade data as a proxy for the presence of productive capabilities.

Empirical evidence indicates that the ECI can outperform other fundamentals—such as education levels, institutional quality or competitiveness rankings—in predicting long-term GDP per capita growth.⁵ The ECI is also used to help explain global differences in income inequality¹⁵ and greenhouse gas emissions per unit of GDP.²⁰

With an ECI of -0.45 on a non-gold basis, Rwanda currently lies near the mid-range of global economies. The key challenge is to translate this position into a broader base of higher-value production. By embedding new knowledge into manufacturing and services, Rwanda can shift from growth driven by volume to growth driven by capability.

Figure 4. Economic Complexity vs GDP per capita (2021–2023)



Countries with higher economic complexity tend to achieve higher income levels over time; points further “up and to the right” combine higher complexity with higher income. See Appendix 2 for methodological details.

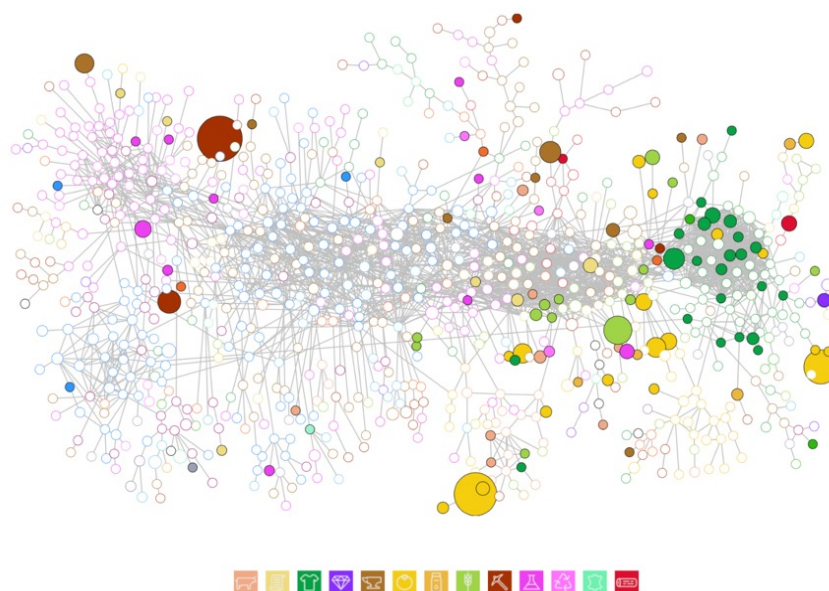
From Diagnosis to Direction

Rwanda's exports indicate what the country already produces competitively. This is the core insight of economic complexity: countries accumulate productive capabilities that shape what they can make today—and what they can make next. Economic complexity therefore helps identify not only a country's current strengths, but also the most feasible pathways for future diversification.

Rwanda's current strengths—including coffee, niobium and tin ores—show revealed comparative advantage ($RCA > 1$). The figure below situates these products within the product space, a network that maps goods commonly co-exported across countries. Products located close to Rwanda's existing exports tend to require similar skills, inputs and infrastructure, making them more feasible entry points.

This mapping helps estimate how feasible it would be for Rwanda to branch into new activities. At present, Rwanda sits near the periphery of the network, but has clear potential entry pathways through textiles, chemicals and plastics.

Figure 5. Rwanda's Core No-Gold Product Space



Products with $RCA > 1$ in Rwanda are highlighted. Proximity on the map reflects similarities in required capabilities; connected nodes represent more feasible next steps. See Appendix 2 for methodological details.

Valuable as they are, these existing niches remain narrow and weakly connected, offering limited pathways into more complex industries. The challenge is to identify products that build on Rwanda's current base while opening routes towards higher sophistication. By quantifying the feasibility of each potential product and combining these estimates with measures of complexity, it is possible to define a cost-effective strategy for structural transformation.

Rwanda's Diversification Frontier

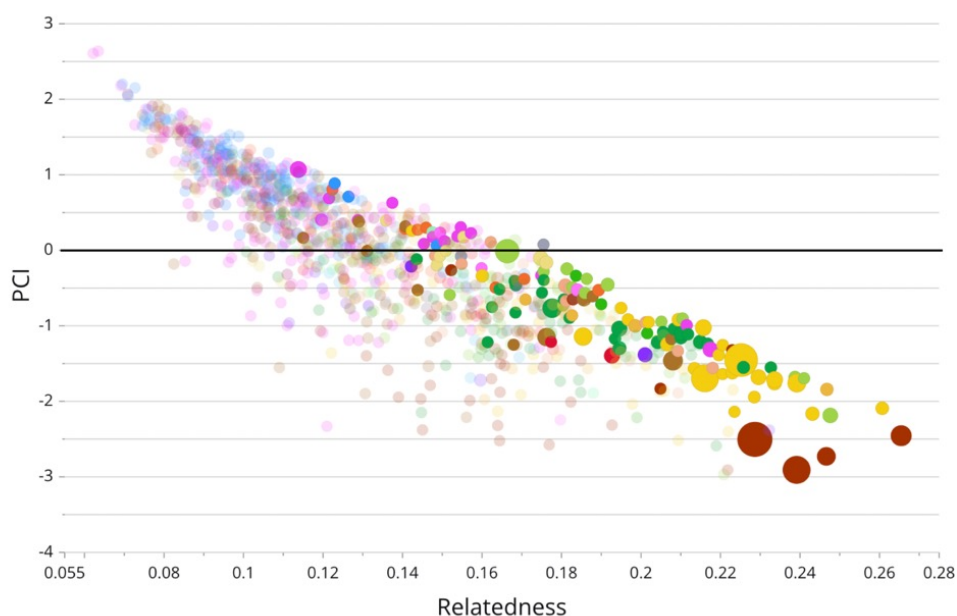
One way to visualise Rwanda's diversification opportunities is through the diversification frontier, which plots each product against two dimensions: relatedness and product complexity. Relatedness^{2,4} measures how strongly a potential product is connected—through shared capabilities—to products that Rwanda already exports. Countries are empirically more likely to enter products with higher relatedness, as these typically require familiar inputs, technologies and supplier networks. Product complexity, by contrast, reflects how knowledge-intensive a product is and the extent to which it can contribute to long-run growth.

Plotted together, the chart highlights a core trade-off between relatedness and product complexity. Relatedness captures proximity to Rwanda's existing capabilities in the product space, which can lower entry barriers through transferable know-how, inputs and suppliers. Product complexity captures how knowledge- and coordination-intensive production is, which tends to increase the scale of capability gains once a product is successfully established.

On the diversification frontier, the lower-right quadrant concentrates feasible, fast moves that can support early industrial growth but deliver modest structural change. The upper-left quadrant concentrates more transformative options that sit further from current capabilities, carry higher

execution risk, and require stronger institutions, more stringent quality systems and deeper investment. The middle quadrants define the upgrade lane: high-relatedness, higher-complexity products often represent the most reliable step-up pathway, while low-relatedness, lower-complexity products can still be difficult because constraints relate to missing complements rather than technical sophistication. These positions indicate probabilities, not guarantees, as feasibility also depends on standards, scale, regulation and access to critical inputs.

Figure 6. Rwanda's Core No-Gold Diversification Frontier



Each dot represents a product. The horizontal axis shows relatedness, indicating ease of entry, and the vertical axis shows complexity, reflecting development potential. Products further to the right are closer to Rwanda's current capabilities; those higher up offer greater long-term gain

Used in this way, the frontier is a diagnostic tool, not a target. It helps sequence diversification so that early wins build the capabilities needed for more advanced entry. Combined with the ECI-OPT model, it becomes a roadmap¹: ECI-OPT identifies a feasible product portfolio, while the frontier supports interpretation and sequencing by linking near-term opportunities to longer-term capability accumulation.

Feasibility in Context: Thresholds That Shape Rwanda's Options

A product may appear feasible on paper but still fail in practice. For Rwanda, two constraints are decisive in determining which diversification opportunities are realistic and viable.

1. Logistics threshold. As a landlocked economy, Rwanda faces high transport costs that can erode margins for low-value or bulky goods. To be commercially viable, potential exports should exceed a unit-value floor of US\$1,654 per tonne, corresponding to the 25th percentile of global export prices².

¹ ECI-OPT sets a target ECI and scores products by the RCA increase needed to reach specialization ($RCA \geq 1$) over the horizon, estimated from an RCA forecast using past RCA and product-space proximity (relatedness/relative relatedness). It then selects the least-cost portfolio meeting the target and plots the selected products on the diversification frontier (relatedness vs complexity) to guide sequencing and policy intensity.

² Derived by computing unit values (US\$/tonne) from BACI's global trade dataset (2023) and setting the

2. Capability threshold. At Rwanda's current stage of development, peer economies typically specialize in products with a Product Complexity Index (PCI) of 1 or below. Prioritizing this range focuses attention on nearer-term entries on the diversification frontier's lower-right quadrant (more feasible, incremental upgrades), while building the routines, supplier networks and quality systems needed to move towards more complex industries in the upper-left quadrant (higher-impact, harder-to-enter activities).

Together, these thresholds help ensure that diversification remains anchored in practical constraints rather than theoretical potential, and that progress reflects genuine accumulation of skills, inputs and institutional readiness.

What's at Stake

If Rwanda advances the 52 near-term products identified in this report—supported by reforms in investment de-risking, skills development, fast-lane regulation and digital infrastructure—the country can shift from concentration-driven earnings, still anchored in a narrow mineral export base, to capability-driven growth. This approach builds the routines and quality systems required to move from lower-right “quick wins” towards higher-complexity, higher-impact entries in the upper-left of the diversification frontier.

This shift would reduce reliance on re-exports and commodities by embedding new technical know-how into production systems and expanding technological capabilities across sectors. As firms gain experience in blending, packaging, quality control and certification, they establish the foundation required for more complex manufacturing in areas such as plastics, electronics and medical devices.

The gains extend beyond export diversification. A more sophisticated production base can improve the quality of industrial jobs, reduce vulnerability to external shocks, and broaden Rwanda's fiscal and innovation space. Moving towards knowledge-intensive production can also support environmental and social objectives by lowering emissions per unit of GDP and expanding participation in growth.

The core question is not only how fast Rwanda grows, but how it grows. By moving from an economy that relies primarily on scale to one that builds and applies capabilities, Rwanda can anchor long-term prosperity in the skills of its firms and workforce and establish a durable pathway towards inclusive and sustainable industrialisation.

Indicator	Baseline	Target*	Why it matters
Economic Complexity (ECI) Core No-Gold**	−0.45	+0.18	Predictor of growth, jobs, resilience
GDP per capita growth (annual)	3.9 %	~5.0 %	Faster compounding, more fiscal space
Emission intensity (kt CO ₂ -eq / bn USD GDP)	551	432	−21.6 % per unit, even without cleaner fuel mix
Inequality (Gini index)	52.3	52.3	Stable distribution of income as complexity rises

Metrics are drawn from the GDP per capita, emissions-intensity, and inequality models described in Appendix 3: Growth, Inequality & Emissions Models.

* Targets assume successful specialization in the 52 products identified through the ECI-OPT model. See Appendix 4 for the full list.

** ECI is standardized relative to the world average (0). Moving from −0.45 (Rwanda, 2023) to +0.18 would lift Rwanda from below to slightly above the world average.

cutoff at the 25th percentile across products; the US\$1,654/tonne figure is that percentile threshold.

Chapter 2

Opportunity Landscape

This chapter applies the ECI-OPT results to identify concrete diversification pathways. It highlights eight flagship products grouped into three strategic clusters that represent progressively deeper capability development: Foundational Capability Upgrades, Adjacent Growth Opportunities and Strategic Innovation Bets. These products build on Rwanda's existing strengths while opening entry points into more complex activities.

Each flagship is assessed using a common set of indicators covering feasibility, capability gains and market potential. RCA indicates whether Rwanda already exports the product competitively. PCI captures the product's knowledge intensity. Relative relatedness measures proximity to Rwanda's existing capabilities. Target ECI growth reflects the expected contribution to export-basket complexity. E&MA imports, top markets and fastest growing markets summarise accessible regional demand. The price range (US\$/t, P10/P50/P90) reports the spread in global shipment-level unit values and indicates the specifications Rwanda would need to meet.

Taken together, these indicators identify products that combine feasible near-term entry with meaningful upgrading. They highlight where Rwanda can compete now, where certification and process improvements can unlock higher-value segments, and where regional demand can sustain industrial learning.

A simple example illustrates the logic. For HS 3208 (nonaqueous paints and varnishes), a relative relatedness score of 0.73 suggests that many required inputs and routines are already in place. Its PCI of 0.59 indicates above-average sophistication, while remaining within reach given Rwanda's current capability base. The model then estimates the increase in RCA needed to reach competitiveness, defined as $RCA \geq 1$. HS 3208 is prioritised because it offers a clear complexity gain with a modest capability gap. The remaining flagships follow the same approach. 3

Flagship 1: HS 3208 – Nonaqueous Paints

Cluster: Adjacent Growth Opportunities

PCI	Rel. Relat.	E&MA imports (US\$M)	Entry SKUs	Key QA/cert	Supply gaps Rwanda can target
0.59	+0.73	191.3	Primers; enamels; waterproof coatings; road-marking paints	ISO 12944; ISO 9001	Industrial coatings; road-marking systems; mid-tier protective paints

Opportunity Overview

Eastern and Middle Africa imported US\$191.3 million of non-aqueous paints in 2023, a 31.5 per cent increase since 2018. Rwanda's domestic market has grown by more than 20 per cent annually between 2019 and 2022, reaching approximately US\$1 million. The product aligns well with Rwanda's current capabilities. Its PCI of 0.59 and relatedness of +0.73 link directly to Rwanda's experience in adhesives (HS 3506) and plastics (HS 3922). Building formulation capacity strengthens dispersion

3 HS codes are Harmonized System product classifications used in trade data; HS 3208 refers to Paints and varnishes based on synthetic polymers or chemically modified natural polymers, in a nonaqueous medium. All trade data comes from the Observatory of Economic Complexity (OEC).

control, viscosity management, batch traceability and testing routines, which can support broader upgrading in chemicals and polymers.

Current Foundations

Rwanda hosts several paint producers; however, current capacity remains well below national demand. AMACO Paints in Gahanga Industrial Park manufactures decorative, automotive and industrial coatings and exports to the Democratic Republic of the Congo (DRC) yet supplies less than 10 per cent of domestic consumption. Other firms, including Maxoline, operate at a similar scale. All rely on imported acrylic resins, pigments, solvents and additives. High input costs reflect fragmented orders, long lead times and the absence of shared bulk storage infrastructure that could reduce unit costs.

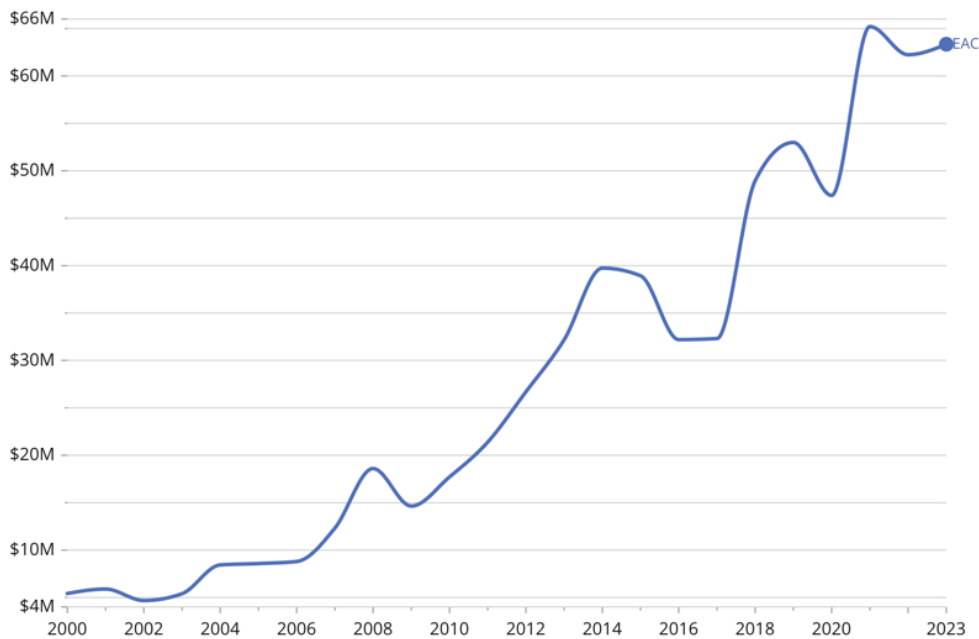
Quality assurance is a primary constraint. Rwanda currently lacks an accredited coatings laboratory, limiting firms’ ability to certify to relevant standards such as ISO 12944 and ISO 9001. Without certification, producers face barriers to accessing procurement markets in construction, roads, utilities and industrial maintenance, which keeps the sector concentrated in lower-specification decorative products, even as domestic demand rises.

Market and Competitiveness Opportunity4

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing importers (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10/P50/P90)
Egypt	32.1	DRC	13.7%	864 / 4,678 / 10,041
Kenya	23.9	Cameroon	12.4%	
Tanzania	19.1	Gabon	11.9%	

Demand in the East African Community (EAC) for non-aqueous paints rose by 63.3 per cent in 2023, supplied largely by the United Arab Emirates, India, Italy and Uganda. This suggests that regional production still covers only a narrow range of specifications.

East African Community (EAC) Imports of Nonaqueous Paints



4 Calculations made with data from the Observatory of Economic Complexity (OEC)

EAC import structures point to clear capability gaps. Kenya's US\$9.1 million market depends on Italy (35 per cent) and Egypt (26 per cent), indicating unmet demand for industrial and automotive coatings that require certified performance. Tanzania's US\$3.6 million import mix from Italy (26 per cent) and China (16.3 per cent) reflects a split between high-specification protective products and low-cost, general-purpose paints. Uganda's US\$7.5 million reliance on India (67 per cent) suggests persistent constraints in standard industrial paints, where batch consistency is essential.

The EAC's 35 per cent Common External Tariff raises land prices for extra-regional suppliers and provides local firms with a cost advantage, provided they meet basic quality assurance requirements and supply reliably. Freight costs and lead times also strongly influence procurement decisions in these segments. Rwanda can enter where responsiveness and consistent batches matter most, including road-marking paints, primers and waterproof coatings. With accredited testing, Rwanda can expand into certified protective coatings for construction, transport and energy infrastructure.

Investment and Capability Roadmap

Rwanda can build competitiveness through a phased approach. In the first year, firms should expand blending and dispersion capacity to 2–3 million litres annually, with an investment of US\$2.5–3.0 million, focusing on primers, enamels, waterproofing and road-marking paints. A shared bulk-input facility in the Kigali Special Economic Zone (SEZ), costing approximately US\$1.5 million, would consolidate imports of resins, solvents and pigments and reduce freight volatility by enabling larger shipments.

Between months 12 and 36, Rwanda should establish an accredited coatings laboratory capable of certifying to ISO 12944 and ISO 9001. Adding colorimetry, accelerated weathering, adhesion and corrosion-resistance testing would enable producers to meet regional technical requirements. With this capability in place, firms can introduce higher-value industrial coatings for infrastructure and construction across the EAC.

From year three onwards, Rwanda should expand distribution into eastern DRC and Bujumbura, supported by regional hubs and tinting centres that shorten lead times and enable on-site customisation. Long-term resin supply agreements would help stabilise input costs and improve competitiveness. At this stage, Rwanda can move into premium protective systems for logistics, industrial and infrastructure projects, shifting the sector towards higher-value, specification-driven production.

Benchmark Case

Crown Paints (Kenya) produces about 3 million litres per month and operates in Uganda, Tanzania and Rwanda. Its scale and quality management systems enable participation in government and private tenders. Basco Paints (Duracoat, Kenya) reports installed capacity of about 22 million litres per year and expanded water-based output by roughly 50 per cent in 2022 with a new emulsion plant. Uganda exported US\$3.88 million of HS 3208 in 2023, mainly to South Sudan (57 per cent), Burundi (28 per cent) and the DRC (14 per cent), illustrating the role of regional demand and distributor networks in scaling production.

Flagship 2: HS 3506 · Glues & Adhesives

Cluster: Foundational Capability Upgrades

PCI	Rel. Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.85	−0.44	138.6	PVAc wood glues; PU sealants; solvent cements	ISO 9001; ASTM D1084; ISO 4587	Construction adhesives; waterproof sealants; packaging hot melts

Opportunity Overview

Adhesives are critical inputs for construction, packaging, furniture and automotive repair. Regional imports reached US\$138.6 million in 2023 and grew by 9.4 per cent annually since 2018, representing one of the fastest expansions among mid-complexity chemicals in Eastern and Middle Africa. Import values have risen faster than volumes, indicating a shift towards higher-performance products such as water-based PVAc, polyurethane sealants and hot-melt systems.

With a PCI of 0.85, adhesives sit above Rwanda's current capability base but remain attainable. Rwanda exported US\$358,000 of HS 3506 in 2022, mainly to the Democratic Republic of the Congo (DRC), confirming early access to regional buyers and the feasibility of cross-border distribution. Building a domestic formulation industry would reduce the import bill, strengthen polymer-processing capabilities and support Rwanda's progression into higher-value speciality chemicals.

Current Foundations

Rwanda has no large-scale adhesive producer, but it has several relevant assets. Polymer-processing firms such as EcoPlastic Rwanda and the new US\$50 million compostable-packaging plant already operate blending and dispersion equipment that could support adhesive formulation. The Rwanda Standards Board has experience in accreditation, traceability and ISO-aligned quality systems, all of which are essential for certification-intensive chemical production. The Kigali Special Economic Zone (SEZ) provides appropriate infrastructure and streamlined customs processes for chemical manufacturing and warehousing.

Important gaps remain. Rwanda does not yet have dedicated formulation equipment, certified testing laboratories aligned with ISO 9001 or ISO 14001, or standardized testing protocols such as ASTM D1084 and ISO 4587. The country also lacks sufficient technicians trained in chemical blending, process control and quality management.

Capabilities developed in adhesives and coatings—including formulation control, batch traceability and routine quality-assurance testing—can provide a foundation for later entry into precision or regulated industries. Medical-device assembly, however, requires ISO 13485-compliant clean rooms that must be physically separated from chemical-processing lines. A single firm may operate in both segments, but production areas, air-handling systems, sanitation protocols and documentation procedures must be fully segregated to meet regulatory requirements.

Market and Competitiveness Opportunity

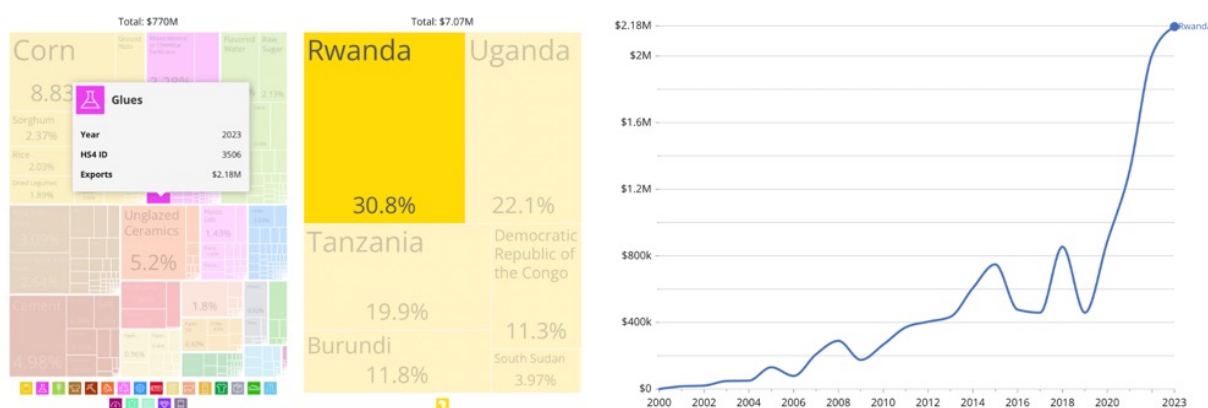
Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing importers (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Kenya	24.3	Ethiopia	13.6%	900 / 2,850 / 10,000
Tanzania	18.8	Tanzania	12.5%	
Ethiopia	15.9	Kenya	11.2%	

Adhesives operate in a two-tier market. Commodity glues from China and the United Arab Emirates sell below US\$2,000 per tonne, while industrial and packaging adhesives from Germany, Italy and India can exceed US\$40,000 per tonne. Median import prices in 2023 reached US\$2,850 per tonne (IQR: US\$900–10,000), indicating substantial demand for higher-performance formulations in East Africa.

Eastern Africa accounts for 65 per cent of regional demand and pays about 23 per cent more per tonne than Middle Africa, reflecting more stringent quality requirements. Kenya, Tanzania and Ethiopia form the high-value core, while Angola and the Democratic Republic of the Congo (DRC) dominate the lower-specification segment.

Rwanda already holds a significant position. Intra-EAC adhesive trade totalled US\$7.1 million in 2023, and Rwanda supplied 31 per cent (US\$2.18 million), ahead of Uganda and Tanzania. Demand for construction adhesives, waterproof sealants and packaging hot-melts favours localised blending, where short delivery times, technical support and product customisation shape purchasing decisions.

East African Community (EAC) Internal Trade: Glues & Adhesives



Investment and Capability Roadmap

A phased approach would enable Rwanda to build capabilities while managing risk. In the first phase, a pilot formulation facility of US\$2–5 million could produce water-based glues and simple solvent cements for packaging and woodworking. The focus would be on blending imported inputs, establishing basic quality testing, and training technicians in polymer handling and batch control.

The second phase would scale output to 1,000–2,000 tonnes per year, with an investment of US\$15–25 million. At this stage, producers could supply regional demand for construction adhesives, grouts and packaging hot-melts. The expansion would add packaging and filling lines, establish basic R&D for bio-based formulations, and work towards ISO 9001 and ISO 14001 certification.

Over the longer term, Rwanda could develop integrated resin and binder production using inputs such as cassava starch or pine resin, aligning with circular-economy objectives and reducing dependence on imported polymers.

Benchmark Case

South Africa provides a strong regional benchmark. Its adhesives sector—anchored by Henkel, Arkema-Bostik, Sasol Chemicals and Permoseal—links local feedstock production with advanced formulation and consistent exports. In 2021, South Africa exported US\$18 million of HS 3506 to SADC markets, driven by high-margin hot-melt and industrial adhesives. Its competitiveness reflects proximity to chemical feedstocks, efficient logistics and targeted incentives, including tax breaks and customs facilitation.

Rwanda does not have South Africa’s feedstock base, but it can replicate elements of the cluster design, quality infrastructure and sustainability orientation. Henkel’s solar-powered operations illustrate a model aligned with Rwanda’s green industry ambitions. Concentrating adhesive production in the Kigali Special Economic Zone (SEZ), integrating it with plastics and packaging firms, and investing in accredited testing would create a cost-efficient, capability-intensive entry pathway aligned with regional demand.

Flagship 3: HS 3819 · Hydraulic Brake Fluid

Cluster: Foundational Capability Upgrades

PCI	Rel Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.86	−0.30	27.2	DOT-3; DOT-4; DOT-4 LV	ISO 4925; FMVSS 116; ISO 9001	Bulk brake fluid for Eastern Africa; premium DOT-4 LV for Middle Africa

Opportunity Overview

Hydraulic brake fluid is a key input in Eastern and Middle Africa’s automotive aftermarket. In 2023, the two regions imported US\$27.2 million across nearly 13,000 tonnes. Although volumes declined from 2022, imports still grew by 6.8 per cent between 2018 and 2023, driven by expanding vehicle fleets, public transport systems and logistics activity.

Rwanda imported close to US\$1 million of brake fluid in 2022. Exports peaked at US\$46,000 in 2018 but have since declined as producers in Kenya, Tanzania and Zambia strengthened their position. With a PCI of 0.86, brake fluid sits above Rwanda’s current ECI but aligns with existing capabilities in blending, quality assurance and packaging, including capabilities associated with lubricants (HS 2710) and adhesives (HS 3506). The product offers a clear import-substitution opportunity and supports regional supply growth that is consistent with Vision 2050 and the 2024–2034 Industrial Policy.

Current Foundations

Local production is anchored by Africa Lubricant Manufacturing Co. (ALMC), a joint venture between Société Pétrolière (SP) and Rymax Lubricants of the Netherlands. Its ISO 9001-certified blending plant in the Kigali Special Economic Zone is estimated to produce 1,500 tonnes annually, blending base oils and additives to global standards (ISO 4925, FMVSS 116). Inputs are sourced from Europe and Asia, while packaging is supplied by local firms such as Inyange Plastics.

This concentrated production base is an asset but also a vulnerability. ALMC demonstrates Rwanda's blending and certification capabilities; however, the country remains dependent on imported base oils and additive packages. Limited bulk storage further increases costs and constrains competitiveness.

Market and Competitiveness Opportunity

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing importers (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Angola	6.1	Djibouti	21.5%	348 / 1,145 / 15,000
Kenya	3.8	Somalia	19.8%	
Tanzania	3.7	DR Congo	13.7%	

Demand divides into two distinct markets. Eastern Africa imported US\$9.3 million and 8,100 tonnes of brake fluid in 2023 and is primarily volume-driven. Average prices were about US\$1,145 per tonne, with large low-cost consignments such as the United Arab Emirates' 4,800 tonnes at US\$348 per tonne. Middle Africa imported US\$17.9 million and 4,800 tonnes and shows a higher-value mix. Average prices were close to US\$3,700 per tonne, and premium shipments from Switzerland and Germany reached US\$15,000 per tonne.

The supplier base is moderately concentrated. Belgium supplied US\$8.1 million, or 29.8 per cent, and the United Kingdom supplied US\$5.8 million, or 21.2 per cent. South Africa supplied US\$3.1 million, or 11.5 per cent. India, the United Arab Emirates and China dominated the lower-priced segment. Belgium, China and South Africa all increased their share between 2018 and 2023, indicating stronger production capability and distribution reach.

For Rwanda, this dual structure offers two entry pathways. Domestically, replacing its US\$1 million import bill is achievable using existing facilities. Regionally, Rwanda can compete in the bulk segment of Eastern Africa on cost and then upgrade into higher-value specifications for Middle Africa, where certification and quality assurance determine market access. The blending and testing infrastructure required for brake fluids also supports adhesives, speciality chemicals and lubricants, which can reduce unit costs across the cluster.

Investment and Capability Roadmap

A phased strategy can scale Rwanda's brake fluid industry. In Phase 1, ALMC could expand with a dedicated 3,000-tonne-per-year line and new bulk storage for glycol ethers, at an estimated cost of US\$1.2 million. This would meet domestic demand and open opportunities in nearby markets such as the Democratic Republic of the Congo (DRC), Burundi, Zambia and Uganda, which together import more than US\$4 million per year.

Phase 2 should focus on premium specifications by securing ISO 4925 Class 6 and FMVSS 116 certifications for advanced DOT 4 LV fluids used in ABS systems and electric vehicles. This would position Rwanda for Middle Africa's higher-value segment, where European suppliers currently dominate. Integrating renewable energy into ALMC operations would align production with Rwanda's green industrialisation objectives.

Policy instruments will play a central role. Customs duty remission on additives, export financing through the Export Growth Fund, and skills partnerships with technical institutes can reduce production costs and accelerate certification.

Benchmark Case

Tanzania offers a relevant regional model. Total Tanzania’s US\$20 million blending plant in Dar es Salaam produces 22,000 tonnes of lubricants and 2,000 tonnes of greases annually, and includes brake fluids for domestic use and export to Kenya, Rwanda and Zambia. The facility benefits from partnerships with oil majors, co-location with port infrastructure for lower-cost input sourcing, and in-house laboratories certified to international standards. Its success demonstrates how early investment in certification, integrated logistics and multi-product blending can secure regional market share.

For Rwanda, the lesson is clear. Integrating brake fluid lines into existing lubricant facilities, prioritising certification and leveraging regional trade corridors can provide a scalable, cost-efficient pathway aligned with SEZ development and AfCFTA export ambitions.

Flagship 4: HS 2922 · Oxygen-Function Amino Compounds

Cluster: Adjacent Growth Opportunities

PCI	Rel Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.70	−0.48	57.3	Glutamic acid; amino salts; basic amino alcohols	WHO-GMP; ISO 9001; ISO 17025	Mid-value amino compounds; EAC pharmaceutical inputs

Opportunity Overview

Oxygen-function amino compounds are used in pharmaceuticals, agro-processing and food fortification. They offer Rwanda a practical entry point into speciality chemicals. With a PCI of 0.70—well above Rwanda’s current ECI of −0.45—the product combines feasible entry with higher complexity. Imports into Eastern and Middle Africa reached US\$57.2 million in 2023, up from US\$40.2 million in 2018, representing a compound annual growth rate of 7.35 per cent. For Rwanda, HS 2922 provides access to a fast-growing regional market and a foundation for progressing towards more sophisticated chemical production.

Current Foundations

Rwanda has begun to build the institutional and industrial base for pharmaceutical production. The Rwanda Biomedical Center manages domestic production of tablets, syrups and infusions, while firms such as Horizon SOPYRWA and Ikirezi Natural Products bring experience in biochemical extraction. These actors demonstrate regulatory oversight and initial capacity in value chains that require quality assurance. However, the country still lacks infrastructure that is most relevant to amino compounds: there are no blending units tailored to this segment, no large-scale synthesis capacity, and limited ISO-certified laboratories. Workforce gaps are also evident, particularly in chemical engineering and advanced quality control. Closing these gaps will require targeted technology transfer, pilot facilities and training programmes aligned with pharmaceutical standards.

Rwanda FDA regulates medicines and APIs, while RSB manages standards and accreditation for chemicals. RBC operates public health laboratories and calibration facilities.

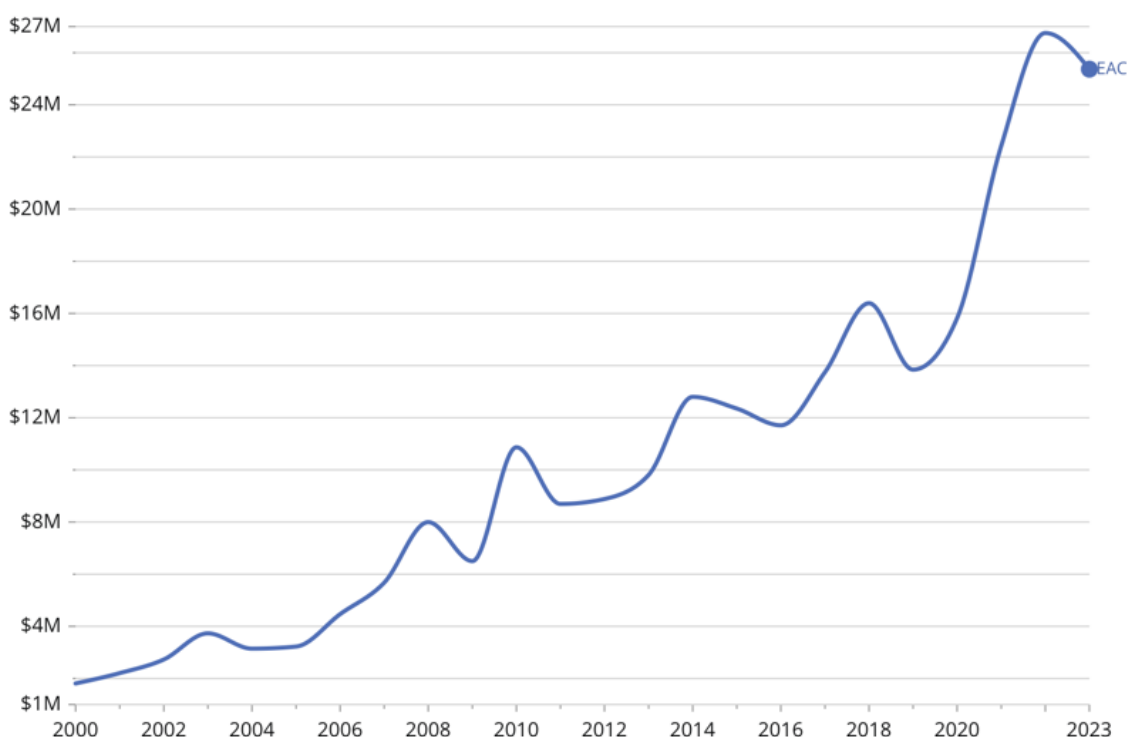
Market and Competitiveness Opportunity

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing importers (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Ethiopia	11.8	Tanzania	20.0%	500 / 1,600 / 45,000
Kenya	9.1	Ethiopia	13.2%	
Tanzania	6.9	DR Congo	12.2%	

In 2023, regional imports of oxygen-function amino compounds reached US\$57.3 million, led by Ethiopia (US\$11.8 million), Kenya (US\$9.1 million), Tanzania (US\$6.9 million), the Democratic Republic of the Congo (DR Congo) (US\$4.4 million) and Sudan (US\$4.1 million). Together, these five markets account for more than 60 per cent of demand. Supply is moderately concentrated: China dominates bulk flows with US\$14.7 million in exports at an average of US\$1,600 per tonne, while India (US\$9.9 million) and Germany (US\$6.9 million) serve mid- to high-value segments, and South Africa (US\$6.6 million) has emerged as a regional supplier.

Trade patterns show a clear price split. Bulk consignments from China are priced between US\$500 and US\$2,000 per tonne, supplying large-volume markets such as Kenya and DR Congo. At the other end, premium shipments from Europe and India reach US\$9,000 to US\$45,000 per tonne, particularly in Ethiopia and Tanzania, where buyers demand higher-quality compounds. The East African Community alone imported US\$25.4 million of HS 2922 products in 2023, more than double its level a decade earlier, underscoring sustained growth in demand across the bloc.

East African Community (EAC) Imports of Oxygen-Function Amino Compounds (2023)



For Rwanda, an immediate entry point is glutamic acid and its salts (HS 292242), where exports grew by 67 per cent to US\$433,000 in 2022 while imports rose by 170 per cent. This dual trend reflects growing domestic consumption and external demand from neighboring markets such as Kenya, Uganda and the Democratic Republic of the Congo (DRC). Establishing a reliable foothold in glutamates would

enable Rwanda to build technical expertise and quality systems and eventually expand into amino alcohols (HS 292219), a smaller but lucrative niche trading at US\$30,000 to US\$70,000 per tonne.

Investment and Capability Roadmap

Rwanda’s entry into oxygen-function amino compounds should build on its current strength in glutamic acid. A pilot-scale facility, requiring US\$2–5 million, could expand production of glutamate and related amino acids while introducing basic amino alcohols, using agricultural residues as feedstock. This first phase would build technical expertise, establish quality systems and test commercial viability.

A second phase—scaling output to 50–100 tonnes annually with an investment of US\$15–25 million—would supply regional pharmaceutical firms and meet WHO-GMP standards. Over time, this set-up could evolve into an integrated biorefinery producing 500–1,000 tonnes and supplying regional value chains.

Parallel infrastructure upgrades will be necessary. Expanding a 1 ktpa blending unit with bulk-packaging lines and ISO-certified laboratories would provide the backbone not only for HS 2922 but also for related product lines such as HS 3506 (glues and adhesives) and HS 3208 (industrial paints).

These steps align with Vision 2050 goals and recent investments such as the US\$150M BioNTech mRNA facility (2024) and the African Development Fund’s US\$12M pharmaceutical technology transfer program (2023).

Benchmark Case

Ethiopia’s pharmaceutical park increased local drug output from 36 per cent to 47 per cent of domestic demand through tax incentives and dedicated infrastructure, although it still relies on imported APIs. Ghana’s US\$321M pharmaceutical park, complete with WHO-GMP-certified laboratories, has begun exporting active ingredients to neighboring markets. Morocco’s US\$20B chemical sector—integrated with European value chains—demonstrates how clustering raw materials, synthesis units and downstream users can drive scale and competitiveness.

Flagship 5: HS 3922 – Plastic Wash Basins

Cluster: Adjacent Growth Opportunities

PCI	Rel Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.55	+0.17	51.6	Standard basins; reinforced basins; premium finish variants	ISO 9001; mold QA; durability testing	Mid-tier household basins; higher-spec items for EAC markets

Opportunity Overview

Plastic wash basins are a fast-growing, mid-tech product with strong regional demand and clear feasibility for Rwanda. Imports into Eastern and Middle Africa reached US\$51.6 million in 2023, an increase of 20.7 per cent over the previous year and almost 50 per cent above 2018 levels—equivalent to an annual growth rate of 8.1 per cent. With a PCI of 0.55 and positive relatedness of 0.17, basins sit just above Rwanda’s current productive frontier. Rwanda already shows a revealed comparative advantage of 0.55, indicating above-average specialisation relative to economies of similar scale.

This combination of early specialisation, rising regional demand and manageable technological requirements makes HS 3922 a strong candidate for expanding local plastics capabilities. Developing production would substitute imports and create an entry point into broader polymer processing, with clear links to adjacent products such as adhesives (HS 3506) and paints (HS 3208).

Current Foundations

A dozen firms form the backbone of Rwanda's plastics sector. EcoPlastic Rwanda transforms post-consumer waste into new products, while multiple companies specialise in pipes, bottles and fittings. Mironko Plastic Industries and SWAN Hydrosol Ltd supply household goods and water-storage tanks, and Depot Kalisimbi Ltd innovates with construction materials made from recycled plastics. However, Rwanda lacks local production of polypropylene and polycarbonate pellets, high-precision moulds manufactured using P20 steel (or higher grades), and sufficient operators trained in injection moulding and thermoforming. These gaps will need to be addressed to enable full-scale basin manufacturing..

Market and Competitiveness Opportunity

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Ethiopia	6.5	Eritrea	84.1%	864 / 4,678 / 10,041
Kenya	5.9	Chad	40.9%	
Tanzania	5.7	Rep. of the Congo	32.2%	

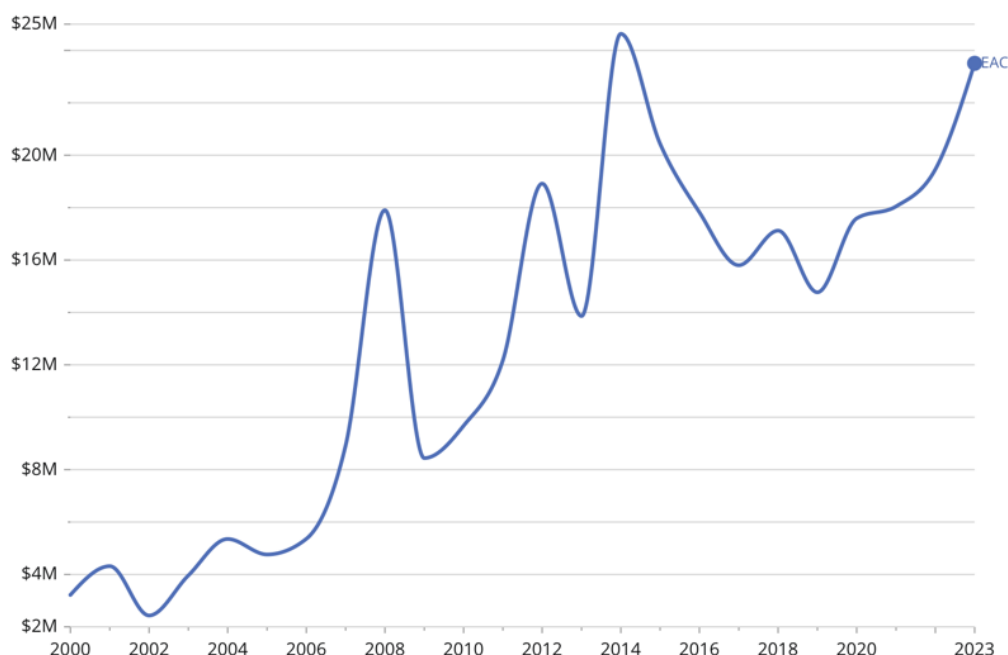
Regional demand for plastic wash basins is concentrated and expanding. In 2023, the leading importers were Ethiopia (US\$6.51 million), Kenya (US\$5.85 million), Tanzania (US\$5.73 million), the Democratic Republic of the Congo (DRC) (US\$5.15 million) and Cameroon (US\$3.64 million). Supply was led by China with US\$26.64 million, or 51.6 per cent of the market, followed by Kenya (US\$4.01 million), India (US\$3.71 million), the United Arab Emirates (US\$3.07 million) and South Africa (US\$1.62 million).

Prices follow a clear two-tier pattern. Weighted averages in 2023 were US\$3,390 per tonne in Eastern Africa and US\$2,195 per tonne in Middle Africa. Shipment-level prices varied widely, with a median of US\$4,678 per tonne and a P10–P90 range of US\$864 to US\$10,041 per tonne. This reflects the coexistence of low-cost commodity lines and higher-specification branded products. The structure supports a tiered strategy that combines standard basins for high-volume markets with upgraded variants that compete on finish, durability and certification.

Within the East African Community, imports reached US\$23.5 million in 2023. Kenya, Tanzania and the DRC accounted for around US\$5–6 million each, equivalent to about 70 per cent of bloc demand. Between 2018 and 2023, these three markets absorbed more than 65 per cent of cumulative imports, with Kenya at US\$30.5 million, Tanzania at US\$21.7 million and the DRC at US\$20.4 million. Rwanda (US\$11.3 million) and Uganda (US\$10.8 million) formed a strong secondary tier, while Burundi, Somalia and South Sudan remained smaller but growing markets.

This geography highlights Rwanda's dual position. The country can reduce its own import bill while capturing nearby demand in the DRC and Uganda, and can gradually build scale to compete with established suppliers in Kenya and Tanzania.

East African Community (EAC) Imports of Plastic Wash Basins



Investment and Capability Roadmap

In the first 12 months, a US\$500,000–750,000 pilot plant—featuring a 450-tonne injection-moulding press, P20 steel moulds and a basic quality control (QC) station—can produce 1,000–2,000 basins per month. Over the next 12–24 months, adding two to three presses, automating cooling and drying lines, and upgrading the QC laboratory to ISO 9001 standards (US\$1–2 million) would raise capacity to 5,000–10,000 units per month. By year three, an investment of US\$5–8 million in additional presses, robotics and in-house pellet recycling capacity could increase output beyond 20,000 units per month, embedding circular-economy practices and reducing unit costs.

Benchmark Case

Kenya's Gjenje Makers began as a small pilot, scaling mixed-plastic recycling into paving bricks and wash basins, quickly reaching 1,000 tonnes per annum and securing county government contracts. In Tanzania, local producers leveraged modern injection-moulding machines and Dar es Salaam's port infrastructure to reach 1.4 million basin units annually and US\$599,000 in exports by 2023. Their experience demonstrates how aligning pilot investments with regional demand and local champions can de-risk expansion and lay the foundation for sustainable, complexity-driven growth in Rwanda. Flagship 6: HS 9018 – Medical Instruments

Cluster: Strategic Innovation Bets

PCI	Rel Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.89	-0.52	658	Thermometers; BP monitors; basic diagnostic devices; simple surgical instruments	ISO 13485; ISO 17025; WHO prequalification	Mid-range diagnostic kits; calibrated basic devices; regional hospital supply gaps

Opportunity Overview

Medical instruments represent one of the most dynamic opportunities in Rwanda’s diversification agenda. In 2023, Rwanda imported US\$18 million of HS 9018 products, a modest 2.1 per cent decline from 2022. Across Africa, the medical devices market reached US\$2.85 billion in 2023 and is expected to grow by 7.1 per cent annually through 2032. The global contract manufacturing market for medical devices already exceeds US\$2.9 billion and is expanding at close to 10 per cent per year.

For Rwanda, the sector aligns with strategic priorities, including hospital network expansion, the pursuit of universal health coverage under Vision 2050 and the goal of building a high-skill, knowledge-intensive industrial base. With a PCI of 0.89, medical instruments sit well above Rwanda’s current ECI of –0.45 and represent a feasible but transformative entry point. Establishing assembly and calibration capabilities would reduce import dependence and position Rwanda within regional supply chains for advanced healthcare technologies.

Current Foundations

Rwanda lacks domestic production of medical instruments but has built critical adjacent capacities. The Rwanda Standards Board hosts ISO 17025–accredited laboratories, while the Rwanda Biomedical Center manages diagnostic and calibration facilities for public-health programs. The University of Rwanda’s College of Science and Technology trains biomedical engineers and technicians, and several private firms maintain hospital equipment. Taken together, these assets provide a technical and institutional base for converting an existing quality-assurance laboratory into a clean-room workshop equipped with autoclaves, torque benches and calibration rigs. This would provide Rwanda with its first end-to-end platform for device assembly and testing.

ISO 13485 requires dedicated clean-room zones with controlled air quality, restricted access and validated sterilisation procedures. These conditions generally preclude co-location with chemical handling or plastics processing and therefore define the operational boundary for regulated manufacturing.

A clean-room platform would build on the quality infrastructure developed through earlier flagships such as amino compounds (HS 2922), adhesives (HS 3506) and paints (HS 3208), and apply it to healthcare—strengthening Rwanda’s progression into precision manufacturing.

Market and Competitiveness Opportunity

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing importers (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Angola	119.0	Chad	35.1%	5,000 / 25,600 / 100,000+
Ethiopia	84.0	South Sudan	25.3%	
Tanzania	83.0	Equatorial Guinea	22.8%	

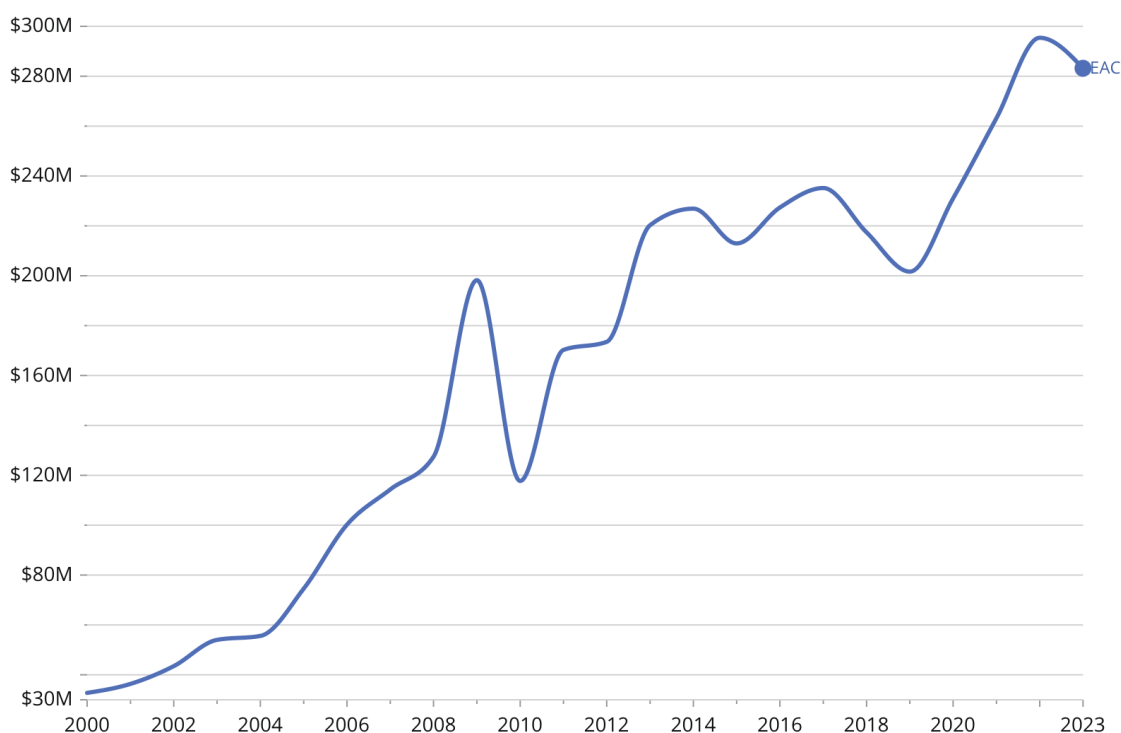
The regional market for HS 9018 medical instruments is large and segmented. In 2023, Eastern and Middle Africa imported US\$658 million and 43,685 tonnes, with compound growth of 7.6 per cent since 2018. Conditions diverged in 2023: demand eased in Eastern Africa but remained resilient in Middle Africa.

Imports are concentrated in Kenya, Tanzania and Uganda, with significant demand in the Democratic Republic of the Congo (DRC), Cameroon and Angola. Two distinct segments drive the market. The first is bulk diagnostic kits and basic devices, typically priced at US\$5,000 to US\$10,000 per tonne. The

second is high-specification instruments, including surgical equipment and portable imaging devices, which can exceed US\$100,000 per tonne. In 2023, the median shipment value was US\$25,600 per tonne, and top-end prices surpassed US\$1.3 million, reflecting the coexistence of commodity and premium products.

Within the East African Community, imports reached US\$283 million in 2023, an increase of 30.3 per cent over five years. Demand has trended upward since 2000, with only brief interruptions during global crises. Capturing even a small share of this market would position Rwanda within a high-complexity manufacturing segment.

East African Community (EAC) Imports of Medical Instruments



Supply is moderately concentrated: China, India and European exporters dominate, while South Africa and the United Arab Emirates act as regional hubs. The top four suppliers control just over half the market, leaving scope for reliable regional entrants. Rwanda's role as host of the African Medicines Agency positions it advantageously, providing regulatory credibility and a platform to compete on quality and compliance.

Investment and Capability Roadmap

Rwanda's pathway into medical instruments requires a phased approach that builds technical capacity while validating market demand. The first stage could convert part of an existing ISO-accredited laboratory into a clean-room workshop with sterilisation autoclaves, torque-testing benches and precision calibration rigs. With an investment of US\$1–2 million, this set-up could produce and calibrate up to 50,000 basic devices per year, including thermometers and blood pressure monitors, while training an initial cohort of 10–15 biomedical technicians.

A second stage, expected within 12–24 months, would add surface-mount technology lines, environmental stress chambers and advanced test benches. This expansion, requiring an additional US\$3–5 million, would double production capacity and allow Rwanda to work towards WHO prequalification, which is essential for larger procurement markets.

By the third year, Rwanda could progress to contract manufacturing by integrating robotic assembly cells and design software, an upgrade costing US\$5–8 million. At this stage, domestic firms would be able to supply regional hospitals and serve global equipment manufacturers that outsource assembly and calibration.

Benchmark Case

Kenya offers a relevant precedent. In 2022, MedSource Labs partnered with GE Healthcare to assemble and calibrate ultrasound probes locally, cutting import costs by 25% and providing same-day technical support for hospitals. South Africa illustrates the role of procurement policy: hospital tender frameworks there reserve a portion of contracts for domestically assembled devices, creating predictable demand and sustaining manufacturing capacity.

These cases point to two levers Rwanda could adapt. Partnerships with global equipment manufacturers can transfer technical know-how and strengthen certification credibility. Public procurement can anchor early demand and help new assembly lines reach scale. Combined, these approaches would enable Rwanda to substitute imports of HS 9018 products and position itself as a regional hub for medical-instrument assembly and calibration.

Flagship 7: HS 8538 – Electrical Power Accessories

Cluster: Strategic Innovation Bets

PCI	Rel Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.99	–0.25	109.8	Stamped metal housings; connectors; terminals; small power-distribution accessories	ISO 9001; ISO 17025 electrical calibration; plating QA	Mid-tier accessories for Ethiopia/Kenya; premium components for Angola/DRC

Opportunity Overview

Rwanda expanded its exports of power-distribution accessories from US\$47,000 in 2021 to US\$690,000 in 2022, but domestic demand still outpaced supply. Imports reached almost US\$3 million in 2022, highlighting a clear production gap. By 2023, East African buyers imported US\$109.8 million of HS 8538 products, reflecting the scale and variety of components sourced from abroad. As the region’s power grid grows from 21 GW to almost 29 GW by 2033, utilities and contractors will require more distribution hardware. If Rwanda builds capacity in this segment, it can narrow its trade imbalance and create an entry point into higher-value electronics assembly.

Current Foundations

Rwanda’s hardware ecosystem is led by firms such as SOLEKTRA Rwanda, which assembles solar-kits and distribution components, and Quincaillerie Beta Ltd, a regional supplier of electrical materials. Brilltech Engineers and SOFARU Hardware import and distribute panel boards, cables and connectors. Despite this base—and ISO 17025-accredited calibration laboratories at the Rwanda Standards Board—no plant currently forges housings, stamps metal contacts, or undertakes electroplating and high-precision assembly. Retrofitting a single workshop bay with a 200-tonne stamping press, a plating line and torque-testing stations could establish Rwanda as a small-scale producer of power-distribution accessories.

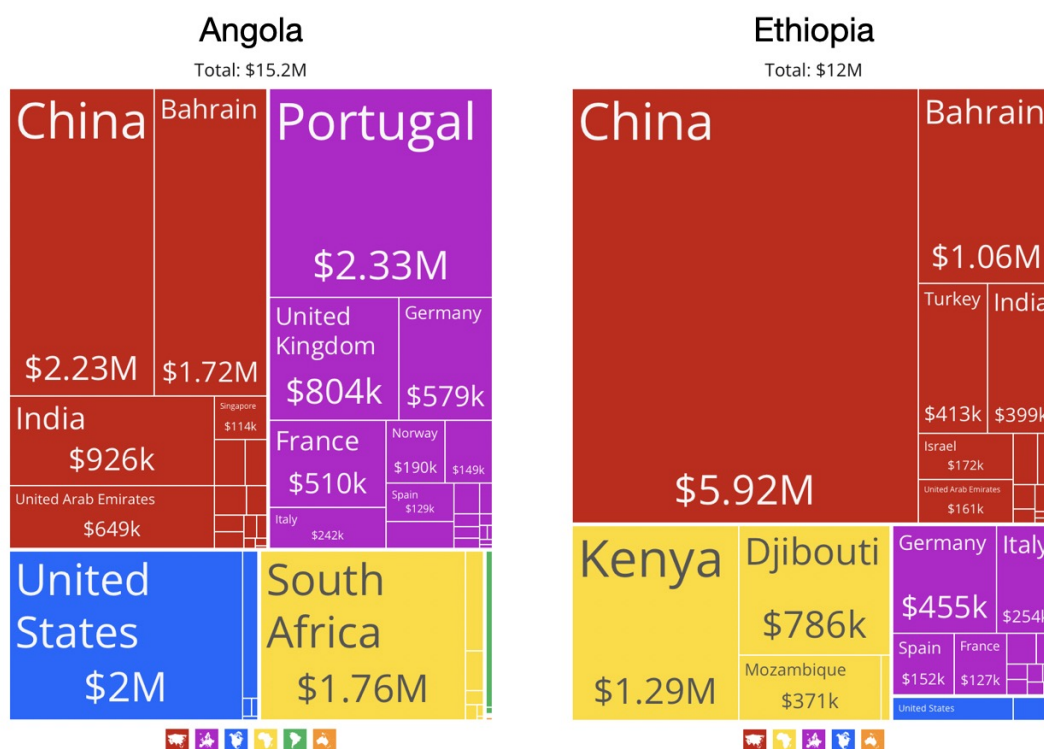
Market and Competitiveness Opportunity

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Angola	15.2	Djibouti	21.5%	8,830 / 18,200 / 20,000+
Ethiopia	12.0	Somalia	19.8%	
DR Congo	11.4	DR Congo	13.7%	

In 2023, Eastern and Middle Africa imported US\$109.8 million of electrical power accessories, an increase of 13.4 per cent from 2022. Eastern Africa accounted for US\$61.1 million at an average price of US\$8,830 per tonne, while Middle Africa imported US\$48.8 million at US\$18,200 per tonne. This gap reflects two tiers: bulk, lower-value goods in Eastern Africa and higher-specification consignments in Middle Africa. After contractions in 2019 and 2020, demand has recovered, with Middle Africa driving much of the recent growth.

Angola illustrates the premium segment. Imports reached US\$15.2 million in 2023 at more than US\$20,000 per tonne, combining bulk shipments from China and Portugal with high-value consignments from Bahrain, Germany, the United Kingdom and South Africa. Ethiopia illustrates the volume-driven tier. Imports rose by almost 59 per cent in 2023 to US\$12 million and 1,600 tonnes, dominated by Chinese mid-tier products with occasional premium shipments from Bahrain and Germany. Djibouti's role as a re-export hub for low-priced, high-volume goods shows how corridor logistics shape competition in inland markets.

Imports of Electrical Power Accessories by Origin (2023)



For Rwanda, these dynamics support a two-track strategy. Cost-efficient production can supply Eastern Africa's mid-price markets, particularly Ethiopia, Kenya and Tanzania. Certified, higher-value accessories can target Angola, the Democratic Republic of the Congo (DRC) and Cameroon, where

utilities and industrial projects pay premiums for quality and reliability. With Africa importing more than 90 per cent of its power accessories, Rwanda can combine domestic import substitution with selective export entry to build a position in both the mid-price and premium tiers.

Investment and Capability Roadmap

A sequenced upgrade pathway is feasible. A pilot investment of US\$0.8–1.2 million would equip a workshop with stamping, plating and torque-testing equipment, enabling production of 100,000–200,000 units annually. A second phase, at US\$2–3 million, would expand plating capacity, add CNC trimming machines and introduce clean-room assembly, increasing output to 500,000 units. At industrial scale, US\$5–7 million could fund robotic handling, automated inspection and polymer overmoulding, raising capacity beyond one million units per year.

Importantly, this pathway can leverage quality and calibration infrastructure already being developed for medical instruments and speciality chemicals, enabling shared use of testing and calibration capabilities across product lines.

Benchmark Case

In 2023, ACTOM Group acquired Schneider Electric's low-voltage assembly plant in Nairobi and expanded it into a regional production site. Within two years, it cut Kenya's import bill for distribution accessories by 30 per cent and secured major contracts with utilities and OEMs. This model combines international technology transfer with local manufacturing scale, supported by strong quality controls and efficient logistics. Rwanda can adapt this approach on a smaller scale by partnering with an established OEM, embedding calibration and plating capacity within Kigali's SEZ, and targeting both domestic substitution and regional exports. Under this approach, HS 8538 could become Rwanda's entry point into higher-value electronics and contribute to the targeted 5 per cent uplift in ECI.

Flagship 8: HS 8473 – Office Machine Parts

Cluster: Strategic Innovation Bets

PCI	Rel Relat	E&MA imports (US\$M)	Entry SKUs	Key QA / certification	Supply gaps Rwanda can target
0.99	–1.06	400	Chassis parts; rollers; gears; small precision components	ISO 9001; ISO 17025 calibration; precision metrology	Mid-range parts for Ethiopia/Kenya; specialty components for high-value Angola segment

Opportunity Review

Office-machine parts—such as chassis, gears, rollers and precision components for printers, copiers and calculators—serve a growing regional market. In 2023, Eastern and Middle Africa imported US\$400 million of HS 8473, equivalent to 14,700 tonnes, at an average price of US\$27,200 per tonne. Trade has grown by 3.6 per cent annually since 2018, despite mid-period volatility, and remains a key input for Africa's expanding digital and office economies. Eastern Africa accounts for 67 per cent of demand, or US\$268 million, while Middle Africa accounts for US\$132 million, with higher unit values.

For Rwanda, this product sits near the upper limit of feasible industrial entry. With a PCI of 0.99, HS 8473 is among the most knowledge-intensive items in the diversification portfolio and is significantly

above Rwanda’s current ECI of –0.45. Securing even a small share of this US\$400 million market would represent Rwanda’s move into precision manufacturing and provide a platform for future electronics assembly.

Current Foundations

Rwanda’s machining ecosystem remains small but evolving. Local workshops in Kigali’s industrial zones produce automotive and electrical parts using basic CNC tools. ISO 17025-accredited laboratories at the Rwanda Standards Board and the Rwanda Biomedical Center provide calibration capacity, while firms such as Brilltech Engineers operate CNC units for enclosures and panels. However, no domestic facility currently produces to $\pm 5\text{ }\mu\text{m}$ tolerances or applies the advanced surface finishing (passivation, electropolishing) required for office-machine components. Retrofitting an SEZ workshop with a three-axis CNC centre, precision broaching and automated inspection rigs could shift Rwanda from general metalwork towards precision parts manufacturing.

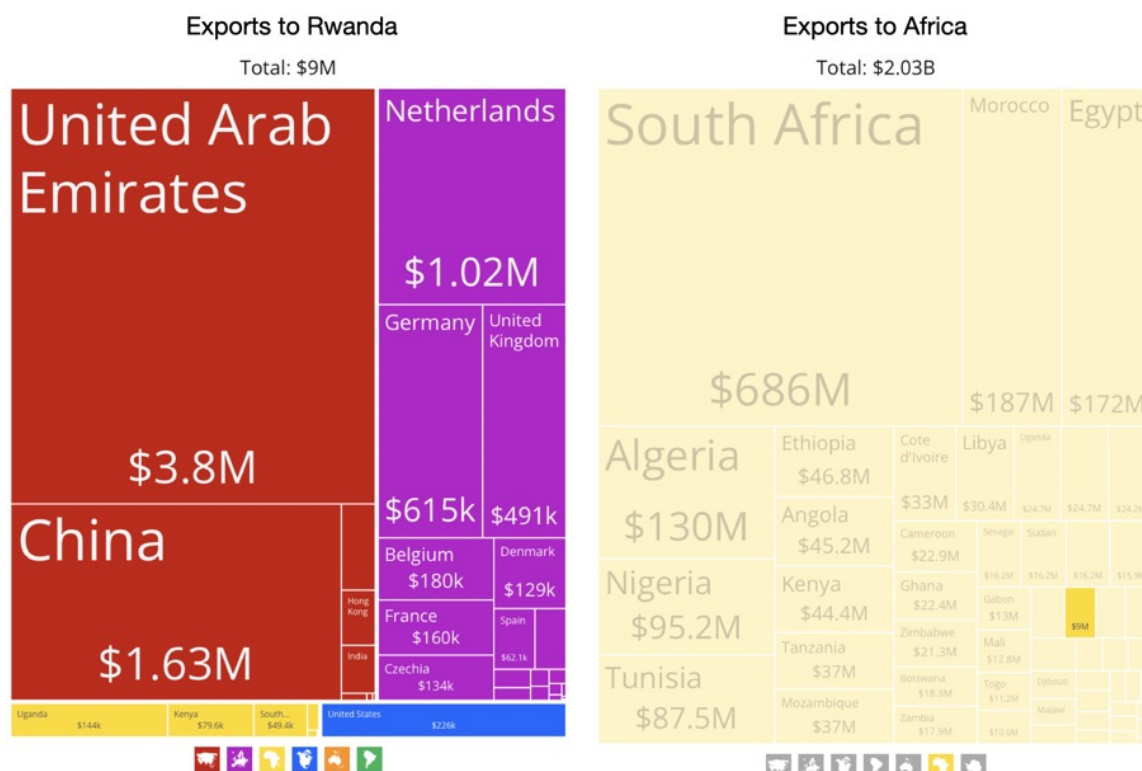
Market and Competitiveness Opportunity

Top importers (E&MA, 2023)	Value (US\$M)	Fastest-growing (2018–2023, CAGR)	CAGR	Price range (US\$/t, P10 / P50 / P90)
Ethiopia	47.0	Gabon	34.8%	200 / 41,000 / 1,000,000+
Angola	45.0	Ethiopia	31.4%	
Kenya	44.0	Cameroon	16.8%	

The regional market for office-machine parts is large and clearly segmented. In 2023, Ethiopia (US\$47 million), Angola (US\$45 million) and Kenya (US\$44 million) were the largest importers, followed by Tanzania and Mozambique at around US\$37 million each. These five markets account for nearly half of Eastern and Middle Africa’s US\$400 million demand. Trade divides into two tiers: bulk commodity parts priced between US\$6,000 and US\$20,000 per tonne, and high-precision components that exceed US\$100,000 per tonne. The median shipment value reached US\$41,000 per tonne, with extremes ranging from US\$200 per tonne for bulk consignments to more than US\$1 million per tonne for specialised modules. This confirms the coexistence of commodity and premium niches.

Rwanda’s position reflects this structure. The country imported US\$9 million in 2023, equivalent to 175 tonnes, at an average price of US\$51,000 per tonne, sourcing mainly from the United Arab Emirates (US\$3.8 million), China (US\$1.6 million) and the Netherlands (US\$1 million). This unit value sits well above the regional weighted average of around US\$25,000 to US\$28,000 per tonne, signalling concentration in higher-specification products. Exports remain small at US\$35,000 but suggest early activity in re-exports or light processing. Across the continent, imports exceeded US\$2 billion, led by South Africa (US\$686 million), Morocco (US\$187 million) and Egypt (US\$172 million).

Imports of Office Machine Parts (2023)



Investment and Capability Roadmap

Rwanda's pathway into office-machine parts starts with a focused pilot that builds precision capability in stages. The first phase requires an investment of about US\$1–1.5 million to install a three-axis CNC machine, broaching tools and an automated optical inspection station in the Kigali SEZ. This equipment could produce roughly 50,000 precision parts per year, supplying local distributors and maintenance firms while demonstrating Rwanda's ability to manufacture to tighter tolerances.

The second phase, expected within 12–24 months, would add a five-axis CNC centre, robotic handling equipment and a passivation line at an additional cost of US\$2–3 million. These upgrades would double annual capacity to around 100,000 parts and enable Rwanda to meet micrometre-level tolerances required for higher-value contracts.

A third phase, with an investment of US\$5–7 million, would introduce clean-room assembly booths, laser-marking systems and small grinding cells. At this stage, annual output would exceed 200,000 units and position Rwanda to supply regional markets under AfCFTA, while establishing a foothold in high-precision manufacturing.

Benchmark Case

In November 2023, French manufacturer Exxelia opened a 10,000-square-metre electronics plant in Mohammedia, Morocco, which includes a 200-square-metre clean room and Industry 4.0 processes for high-reliability modules. The facility shows that clean-room assembly and certified quality systems can operate competitively in Africa while supplying global original equipment manufacturers.

Tunisia provides a second precedent. Its established electronics manufacturing services cluster exports sub-assemblies and finished electronics to Europe, and the national industry registry lists around 340 firms with more than 10 employees in electric and electronic manufacturing, highlighting the sector's depth.

These cases show that Africa can sustain advanced electronics manufacturing and integrate into global value chains. For Rwanda, adapting this model by pairing CNC machining and calibration with AfCFTA-enabled regional exports can reduce the country’s high import bill for office-machine parts and create a viable foothold in precision manufacturing. This pathway is consistent with the projected 4.9 per cent ECI growth contribution.

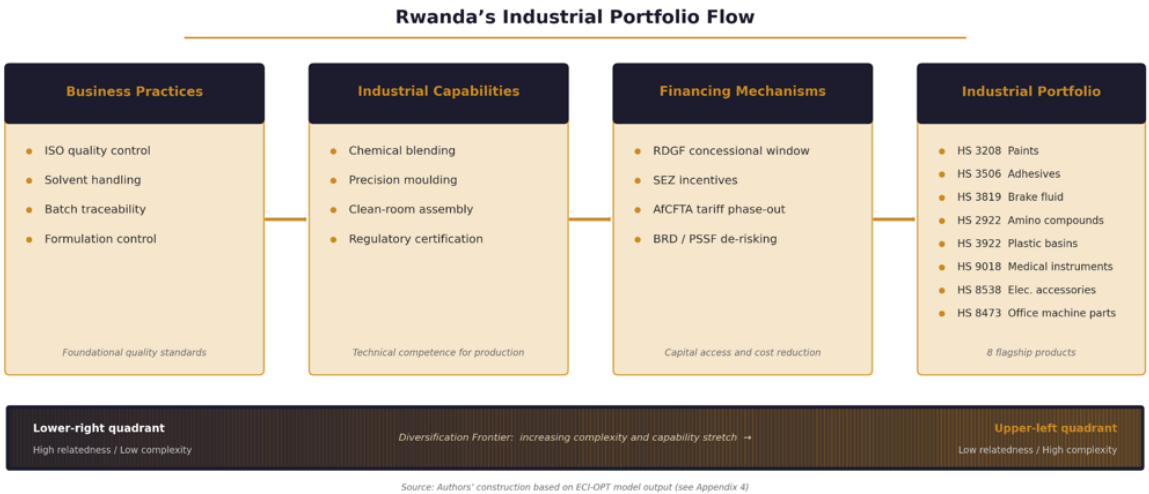
How the Portfolio Works Together

The model relies on shared assets and know-how that transfer across sectors in a pattern consistent with the diversification frontier: capabilities built in more feasible activities can be reused to unlock harder, higher-upgrading entry points. Chemical formulation and blending—including ISO-accredited quality control, solvent handling and batch traceability—create routines that spill over into paints, adhesives and fragrances. This does not imply that all chemicals are “low-entry” or that fragrances are inherently “high-entry”. Rather, it reflects that formulation routines are broadly reusable and are often closer to existing capabilities than regulated manufacturing.

As these standards take root in Rwanda’s testing laboratories and regulatory agencies, plastics firms can adopt them for extrusion and moulding, strengthening precision, calibration and process control. These disciplines, in turn, underpin the highest-entry parts of the portfolio, where certification and compliance are binding. Clean-room routines, measurement discipline and documentation systems developed through plastics and coatings make entry into medical devices and electrical power accessories credible, with medical devices typically among the most demanding activities, given the extent of regulatory requirements.

Financing follows the same ladder. Early formulation wins can justify grants and SME lending for blending units; mid-tech plastics and coatings can attract private capital as processes become standardised and cash flows more predictable. High-risk pilots in medical devices and precision components become investable once earlier stages reduce technology-transfer risk and Rwanda can credibly test, certify and enforce standards. Within AfCFTA’s tariff schedule, which phases out intra-African duties on manufactured goods over 5–10 years, these eight flagships become more competitive regionally, while the RDGF’s concessional window offsets early capital costs that the tariff timeline alone cannot resolve.

The result is a linked portfolio in which each flagship reinforces the next. Chemicals build formulation and quality routines that support plastics; plastics deepen process control that anchors coatings; coatings enable clean-room practices; and clean-room disciplines open pathways into electronics and medical devices. Each step raises the returns to prior investments in equipment, standards and skills, increasing complexity by connecting projects so that capability gains compound across the system.



Chapter 3

Making It Happen

Implementing this strategy requires more than selecting products. Rwanda must run an execution system across the full investment cycle, mobilising blended capital through the Rwanda Development Guiding Fund, fast-tracking strategic projects via a fast-lane regulatory pathway, embedding technical expertise through an Open Economy for Knowledge Workers, and enforcing accountability through a real-time industrial delivery system.

Rwanda Development Guiding Fund

One lesson from China's industrial ascent is the role of guiding funds: blended public–private vehicles that crowd in capital for priority industries without distorting markets. Zhongguancun, often called “China's Silicon Valley”, was not built through subsidies or showcase factories, but through disciplined funds that signalled national priorities, shared risks transparently, and attracted sophisticated investors. This model showed that patient, risk-sharing capital can accelerate capability building while keeping market incentives intact.⁵

In Rwanda, the Rwanda Development Guiding Fund serves the same purpose. It operates as a permanent blended-capital platform that pools public and private resources on *pari passu* terms. Government seed capital shares upside and downside equally with private investors. Anchored by BRD and RDB, the fund aims to mobilise US\$100–150 million by combining US\$50 million of public seed financing with contributions from development partners, diaspora bonds and private investors. By targeting mid-sized industrial projects requiring between US\$2 million and US\$20 million, the RDGF fills a critical financing gap where promising ventures often stall between pilot and scale.

This design also complements—and improves on—Rwanda's existing recovery-era financing toolkit. The BRD/MINECOFIN Economic Recovery Fund and related facilities, including PSSF-derived instruments, have been effective at stabilising firms and extending liquidity through credit lines, guarantees and risk-sharing. However, these approaches are predominantly debt-oriented and best suited to working-capital needs or de-risking bank lending. The RDGF adds the missing equity-style layer needed for industrial upgrading: patient capital that can absorb early uncertainty, finance capability-building investments, and remain invested through the scale-up phase.

Governance arrangements are central to the fund's credibility. An independent investment committee, staffed with financial professionals, would operate at arm's length from day-to-day politics and remain accountable to a private-sector advisory council. Government contributions would be repaid at cost plus a capped premium, which maintains fiscal discipline and allocates most upside to investors. The RDGF is therefore a structured equity vehicle designed to align incentives and crowd in private capital, not a subsidy instrument.

The RDGF mandate is to strengthen industrial investment while avoiding future fiscal pressure. The fund's design reflects this requirement. Government participation is capped at a predetermined share of each transaction, and public seed capital is deployed only on *pari passu* terms with private investors. Portfolio-level risk controls limit the Treasury's maximum exposure and prevent public capital from absorbing open-ended downside risk. As investments mature, government returns are recycled to preserve the fund's capital base rather than create new recurrent obligations. This structure keeps the instrument fiscally neutral while allowing private investors to remain the principal drivers of project selection, risk assessment and scale-up.

⁵ More details on guiding funds can be found in: *The Infinite Alphabet: And the Laws of Knowledge* by César A. Hidalgo (Penguin, UK) (2025).

Rwanda's own experience shows the feasibility of this model. The Rwanda Innovation Fund mobilised private investment in high-risk tech SMEs through structured guarantees backed by the AfDB. BRD's growth facilities unlocked US\$30 million in SME lending by reducing bank-level credit risk. The Export Growth Fund blended concessional credit with partial guarantees to help firms expand production and reach new markets. Yet each initiative has been debt-focused, sector-specific or temporary. None created a systemic, equity-style mechanism to finance industrial upgrading at scale. The RDGF fills that gap.

Other African precedents confirm that this approach is viable. South Africa's Industrial Development Corporation has long used equity, quasi-equity, debt and guarantees to support industrialisation while maintaining commercial discipline. In Morocco, the Fonds Mohammed VI pour l'Investissement operates as a public-private investment platform that directs capital to strategic sectors through thematic and infrastructure-focused funds. These institutions show that blended-capital platforms in Africa can combine clear mandates, rigorous governance and developmental objectives in ways that mirror the RDGF's design.

The RDGF would extend this model in Rwanda by linking directly to the country's complexity roadmap. Early wins in chemicals and adhesives justify follow-on grants and loans. Mid-tech investments in plastics and paints attract private investors because cash flows are predictable. High-value pilots in medical devices and machine parts become investable once early phases demonstrate Rwanda's ability to regulate and certify production. In this way, the fund becomes the financial backbone of the industrial strategy rather than a stand-alone initiative.

A permanent guiding fund would move Rwanda beyond ad hoc programmes and isolated guarantees. The RDGF would translate strategic intent into investable projects and enable the eight flagship products and their clusters to scale from concept to competitive industries. If implemented effectively, it would position Rwanda as an early adopter of blended equity at scale in support of higher-complexity production.

Open Economy for Knowledge Workers

Capital will not raise complexity without people who know how to build, certify and export. Rwanda needs an open, targeted talent regime to deliver the flagship products in chemicals, plastics, power accessories, medical instruments and precision parts that anchor this strategy.

Rwanda has already built part of this platform. Carnegie Mellon University Africa (CMU-Africa) is a graduate campus in Kigali Innovation City that offers MS programmes in information technology, electrical and computer engineering, and engineering AI. It is complemented by other talent pipelines, including the African Institute for Mathematical Sciences (AIMS) and the Rwanda Coding Academy, which have helped pioneer data science and AI training locally. CMU-Africa reports more than 320 students, over 870 alumni, and a 94 per cent employment rate within one year of graduation. Together, these cohorts supply engineers and data professionals who can staff quality laboratories, automation projects and regulated production lines.

Rules exist for skilled entry. A 2022 ministerial order establishes the Occupations in Demand List and sets procedures for labour-market testing and classification. The order recognises East African Community (EAC) Common Market provisions for liberalised occupations. Rwanda's V4 job-search visa allows foreign professionals whose skills appear on the list to enter for up to 90 days while they secure employment. These instruments define a legal pathway for targeted technical recruitment.

Demand for experienced specialists is real. According to the National Bank of Rwanda, firms with foreign private capital employed 59,916 people in 2023, including 1,501 foreign specialists across short- and long-term roles. This small but critical cohort helps commission plants, install quality systems and train local teams.

To reduce hiring friction for technical roles, Rwanda should strengthen HEC's foreign degree-equivalence system by setting a 10-day service level for evaluations, publishing monthly processing statistics, and expanding the list of pre-approved accredited institutions for priority occupations. Faster and more predictable decisions would reduce onboarding delays for engineers, chemists and technicians in flagship industries.

Rwanda can now pull these pieces into a single, execution-oriented framework:

1. **Critical Skills Track.** Publish an annual Occupations in Demand List aligned to the industrial portfolio and its HS codes. Establish a 10-day service-level agreement for permits and accreditation decisions for listed roles. Report quarterly metrics on approvals, rejections and processing times.
2. **Kigali Skills Exchange.** Create a placement hub at CMU-Africa that matches firms in flagship product lines with graduate engineers and mid-career specialists. Require participating firms to submit training plans that pair each foreign hire with two Rwandan understudies. Track training completion and wage outcomes.
3. **Diaspora Technical Assignments.** Launch 6- to 18-month secondments for Rwandan professionals abroad to lead factory start-ups, certification audits and automation retrofits. Use the Occupations in Demand channel for expedited processing and publish placement data linked to the flagship pipeline.
4. **Regulated-industry anchors.** Tie fast-tracked talent to projects that expand Rwanda's regulatory credibility. The Kigali mRNA facility shows how frontier production requires high-skill teams. Prioritise similar anchors in devices and speciality chemicals and link them to structured on-the-job training.
5. **Measurement.** Use the current count of 1,501 foreign specialists in firms with foreign private capital as the baseline. Track placement shares by sector, reduce permit-processing times, and increase the ratio of certified technicians to total plant staff. Publish results on the same public dashboard that tracks investment and exports.
6. **HEC equivalence services:** Commit to a 10-day service level for foreign degree recognition, publish monthly metrics, and pre-approve equivalence for degrees from accredited institutions in priority occupations.
7. **This design aligns Rwanda's industrial plan with a defined talent pipeline.** It builds on existing universities, immigration rules and accredited laboratories, and then adds speed, transparency and placement discipline. The result is faster learning by doing and a steeper trajectory up the capability ladder that underpins long-term competitiveness.

Fast-Lane Regulation — Unblocking Projects in 120 Days

Rwanda can move strategic plants from commitment to commissioning within four months by converting today's service charters and digital systems into a single, time-bound approvals track. The baseline already favours speed. Rwanda ranks 38th globally on the World Bank's Doing Business 2020 index, with seven days to register property, 30 days to obtain an electricity connection in the standard case, and a 97-day benchmark to obtain a construction permit for a warehouse. The Rwanda Development Board issues investment certificates within two working days under Article 12 of the 2021 Investment Code and its service charter, and company registration services are delivered within six working hours by the Office of the Registrar General. These are assets the fast-lane should institutionalise.

The fast-lane would formalise a single project officer for each strategic investment. RDB already assigns focal persons to approved projects. Embedding this role into a 120-day pathway means the project officer triggers all required approvals on day zero, including environmental clearance, land and construction permits, utility connections, standards compliance and customs facilitation. The approach

does not create new procedures. It sequences existing ones in parallel and enforces the service-charter timelines already published by the Government.

Environmental approvals set the critical path in many industrial projects. Current law and practice support predictable timelines. RDB's service charter indicates 15 days for EIA clearance and 45 days for the EIA certificate once a complete file is submitted. Recent guidance referencing Ministerial Order 001/2019 also indicates that RDB should analyse a submitted EIA report within 20 working days, with an additional 15 working days when a public hearing is required. The fast-lane should commit to these timelines, publish when the clock starts, and stop it only when the applicant is formally asked for additional information.

Transparency & Accountability – Turning Strategy into a System

Industrial transformation requires disciplined execution. Rwanda has built strong performance systems through Imihigo contracts, the Rwanda Governance Scorecard, and structured monitoring of Vision 2020 and Vision 2050. These tools have strengthened coordination and accountability across ministries. Industrial upgrading adds new requirements because production, certification, skills and investment decisions must move together for firms to scale. Rwanda's digital infrastructure provides the backbone for real-time monitoring.

The Irembo platform, documented in the Digital Foundations section, has processed over 25 million applications through 7,000 agents, with transactions above US\$300 million, reducing service times to under 24 hours. These capabilities demonstrate that Rwanda can manage high-volume, time-bound processes across agencies.

Although regulatory agencies use digital systems, firms still encounter fragmented workflows across standards, environmental approvals, utilities and SEZ operations. Rwanda should gradually harmonise these platforms into a single approvals interface with shared case IDs, integrated document repositories, and synchronised service-charter timelines. A unified system reduces coordination costs and helps ensure that fast-lane projects move through all agencies with consistent processing.

To translate industrial strategy into delivery, Rwanda can formalise a Transparency & Accountability System for Industrial Policy. The system would track strategic projects and flagship products through four functions:

1. Real-time dashboard consolidating permitting, land and utility connections, certification cycles, investment inflows, and export volumes. Data would flow from Irembo, RSB's ISO 17025-accredited laboratories, and SEZ systems already described in this report.
2. Quarterly industrial reporting that publishes progress on production, jobs and exports for adhesives (HS 3506), paints (HS 3208), plastics (HS 3922), brake fluid (HS 3819), medical instruments (HS 9018), and electrical accessories (HS 8538). This mirrors the evidence-based monitoring used in Vision 2050 and the 2024–2034 Industrial Policy.
3. Performance compacts for agencies responsible for standards, permitting and SEZ operations, modelled on Imihigo and linked to fast-lane regulatory privileges or budget disbursements.
4. Independent verification reviews conducted with business associations and technical institutes, following global practice in structured delivery units.

Benchmark experience supports this model. Malaysia's PEMANDU used weekly reporting, delivery labs and open dashboards to accelerate reforms in priority sectors. Morocco's Industrial Acceleration Plan applied public scorecards to track progress in industrial ecosystems and guide investor decisions.

By making implementation visible and measurable, Rwanda can align capital, regulation and skills across agencies. A transparent delivery system signals predictability to investors and helps ensure that industrial policy shifts from intent to operational performance.

What Success Looks Like

Rwanda's industrial transformation depends on measurable shifts in productive capacity, technological capability and export composition. Success means new factories operating in priority sectors, firms creating high-quality industrial jobs, and a sustained rise in exports of more complex products that broaden the national knowledge base and strengthen Rwanda's role in regional and global value chains.

A credible monitoring and evaluation framework should track how capabilities, production and exports evolve across the flagship portfolio. Capability indicators include the number of ISO-accredited tests completed each quarter, the number of firms achieving ISO 9001 or ISO 13485 certification, and the pool of technicians trained in chemical processing, moulding and precision assembly. Production metrics should monitor annual output in strategic products, the share of inputs sourced regionally, and line-level defect rates, which together indicate whether firms are scaling and whether quality systems are maturing.

Export performance provides a complementary signal. Rising export values, stronger RCA in each flagship line, and growth in the number of EAC and AfCFTA markets reached will show whether new capabilities are translating into market presence. Regulatory performance must be monitored with equal discipline. Permit-processing times, the share of strategic projects that meet the 120-day fast-lane benchmark, and adherence to published service-charter timelines will indicate whether the regulatory environment is enabling investment. Capital mobilisation should also be tracked. The volume of financing deployed through the Rwanda Development Guiding Fund, the private-to-public leverage ratio, and the number of projects reaching commercial operation will show whether blended finance is functioning as intended.

Publishing these indicators on a quarterly dashboard, owned by MINECOFIN, will provide a consistent basis for assessing progress towards the targeted improvement in Rwanda's Economic Complexity Index and will help identify bottlenecks early.

Operationally, success will appear as shorter approval timelines, faster integration of skilled technical workers into firms, and higher domestic value addition in adhesives, paints, plastics and medical devices. Regular publication of data on investment flows, production volumes, job creation and export outcomes will reinforce accountability and signal to investors that commitments are being delivered.

At the system level, Rwanda will succeed when capital mobilisation, talent attraction, regulatory efficiency and monitoring platforms operate as a single delivery mechanism. This alignment marks the shift from strategic planning to consistent execution. Internationally, Rwanda will be recognised not only as an investment-friendly location but as an emerging manufacturing hub capable of turning industrial policy into real diversification, technological upgrading and inclusive growth.

Conclusion & Next Steps

Rwanda's industrial strategy sets a clear ambition to move from a narrow export base to a more sophisticated and technologically capable economy that creates high-quality jobs and sustained growth. The country already has strong foundations, including a second-place ranking in Africa and a 38th-place global ranking in the World Bank's Doing Business 2020 report. The next step requires disciplined execution built on data, coordination and clear accountability.

The Rwanda Development Guiding Fund can mobilise blended capital for mid-sized manufacturing ventures. These ventures need a fast-lane regulatory track that clears strategic industrial projects in under 120 days. Firms also need reliable access to technical talent, which requires a national framework that integrates visa reform, diaspora placement and targeted skills pipelines. Once projects begin operating, a real-time industrial dashboard should track jobs, exports and investment flows so that policymakers and investors can monitor progress and identify bottlenecks.

Rwanda now needs to shift from individual reforms to system performance. The instruments already exist: capital, regulation, talent and monitoring. Their impact depends on how tightly Rwanda aligns them. Financing without regulatory certainty leaves projects stuck. Rapid approvals without skilled workers reduce competitiveness. Talent programmes without monitoring do not show results. Only a unified execution system can convert national strategy into delivery.

Success will depend on measurable outcomes. These include factories breaking ground on schedule, precision parts entering production, and exports rising in the trade data. For citizens, the proof will come through more high-quality jobs and broader opportunities to gain new skills. Execution will determine whether Rwanda's industrial ambition delivers sustained and inclusive growth.



Appendix

SECTION GLOSSARY



	Animal Products
	Vegetable Products
	Animal and Vegetable Bi-Products
	Foodstuffs
	Textiles
	Foodwear and Headwear
	Plastics and Rubbers
	Animal Hides
	Wood Products
	Mineral Products
	Metals
	Stone and Glass
	Precious Metals
	Machines
	Chemical Products
	Paper Goods
	Weapons
	Transportation
	Instruments
	Miscellaneous
	Arts and Antiques
	Unspecified

1 Data Sets

This analysis draws on two main datasets:

Domain	Source	Coverage	Granularity
Trade flows	OEC (downloaded Feb 2025)	2020–2023	HS4 (1,223 products)
National accounts, human capital, and population	WDI (downloaded Feb 2025)	1995–2023	Country-year

To ensure reliable complexity estimates, only products with at least US\$500 million in global trade across 2020–2023 were included. This filter removes extremely rare items that may distort product rankings.

2 Complexity & Relatedness

2.1 The Logic of Economic Complexity

Economic complexity measures the knowledge embedded in a country's export structure. The more diverse and sophisticated the products a country exports, the more productive knowledge it is understood to possess.

The Economic Complexity Index (ECI) assigns each country a score based on the average complexity of its specialised exports. The Product Complexity Index (PCI) assigns each product a score based on how exclusively it is exported by complex economies. This logic is recursive: complex economies export complex products, and complex products are exported by complex economies.

The mathematical foundation of ECI and PCI lies in the eigenvectors of matrices constructed from Revealed Comparative Advantage (RCA) data. A country is considered specialised in a product if it exports that product with $RCA > 1$. This binary specialisation matrix (M_{cp}) is the basis for calculating both country and product complexity.

Formally:

Let the complexity K_c of a location c (e.g. a country or city) be K_c , and the complexity K_p of an activity p (e.g. a product or industry) be K_p . Let M_{cp} be a binary specialisation matrix summarising the activities present in each location. M_{cp} is defined as 1 when a location's output in an activity is larger than what is expected for a location of the same size and an activity with the same total output. This is computed using an indicator of specialisation known as a location's Revealed Comparative Advantage (RCA), R_{cp} .

That is,

$$M_{cp} = 1 \text{ if } R_{cp} \geq 1, \quad \text{where} \quad R_{cp} = (X_{cp}X_c)/(X_cX_p),$$

where X_{cp} is the volume of activity p in location c (e.g. exports of product p by country c) and

$$X_c = \sum_p X_{cp}, \quad X_p = \sum_c X_{cp}, \quad \text{and} \quad X_{\square} = \sum_{cp} X_{cp}.$$

Following this notation, the Economic Complexity Index is defined as a metric using the following two conditions:

(i) The complexity of a location K_c is a function f of the complexity K_p of the activities present in it M_{cp} ,

and,

(ii) The complexity of an activity K_p is a function g of the complexity K_c of the places where that activity is present M_{cp} .

This logic is equivalent to the following mathematical map

$$\begin{aligned} K_c &= f(M_{cp}, K_p), \\ K_p &= g(M_{cp}, K_c), \end{aligned}$$

where f and g are functions to be determined.

These maps imply that measures of the complexity of economies, or of economic activities, are solutions to self-consistent equations of the form:

$$\begin{aligned} K_c &= f(M_{cp}, g(M_{cp}, K_c)), \\ K_p &= g(M_{cp}, f(M_{cp}, K_p)), \end{aligned}$$

which on many occasions can be reduced—or approximated by—a matrix equation of the form:

$$\begin{aligned} K_c &= M_{cc'} K_c, \\ K_p &= M_{pp'} K_p \end{aligned}$$

The solution to these equations are the metrics of economic complexity for locations (K_c) and activities (K_p). Technically, these equations imply that metrics of economic complexity are the eigenvectors of matrices connecting similar economies ($\mathcal{M}_{cc'}$) or similar activities ($\mathcal{M}_{pp'}$).

These equations also tell us that complexity measures are relative since the complexity of a location, or an activity can change because of changes in other locations or activities (other rows or columns in the $\mathcal{R}_{cp}$ or $\mathcal{M}_{cp}$ matrix).

In the above framework, the Economic Complexity Index of a location, or ECI, is defined as the average of the Product Complexity Index, or PCI, of the activities present in that location. Similarly, we define the Product Complexity Index of an activity, or PCI, as the average Economic Complexity Index, or ECI, of the locations where that activity is present. That is, f and g are simple averages and the ECI formula is the solution to the system of equations:

$$\begin{aligned} K_c &= \frac{1}{M_c} \sum_p \mathbb{1}_{cp} M_{cp} K_p, \\ K_p &= \frac{1}{M_p} \sum_c \mathbb{1}_{cp} M_{cp} K_c, \end{aligned}$$

where $M_c = \sum_p \mathbb{1}_{cp} M_{cp}$ is the number of activities (or diversity) of a location and $M_p = \sum_c \mathbb{1}_{cp} M_{cp}$ is the ubiquity of an activity (number of locations where it is present). Putting the second equation in the first one and solving the system is equivalent to diagonalizing the following matrix:

$$M_{cc'} = \frac{1}{M_c} \sum_p \mathbb{1}_{cp} \frac{M_{cp} M_{c'p}}{M_p}.$$

Since Economic Complexity is a relative metric, the results are usually normalized using a Z-transform. That is:

$$ECI = \frac{K_c - \text{mean}(K_c)}{\text{std}(K_c)},$$

$$PCI = \frac{K_p - \text{mean}(K_p)}{\text{std}(K_p)},$$

where $\text{mean}(K)$ and $\text{std}(K)$ are respectively, the mean and the standard deviation of K .

The economic and product complexity indexes are also known to minimize the equation¹⁵:

$$\sum_c \sum_p M_{cp} (ECI_c - PCI_p)^2$$

And to be an optimal metric of distance to the technological frontier for log-super modular matrices³³.

2.2 Relatedness and the Product Space

Diversification depends not only on what is valuable, but also on what is feasible given a country's current capabilities. This is where relatedness comes in.

Relatedness measures the proximity between a country's current exports and a target product, based on how frequently pairs of products are exported together globally. This proximity is calculated using the product space: a network that connects products with similar input requirements.

Formally:

We estimate relatedness by summing the proximity of a product to all products in which Rwanda is specialised and dividing that by the sum of the proximity between that product and all other products. That is, if the proximity between two products p and p' is $\phi_{pp'}$, then the relatedness between an economy c and an activity p is given by:

$$w_{cp} = \frac{\sum_{p'} M_{cp'} \phi_{pp'}}{\sum_{p'} \phi_{pp'}}$$

Where $M_{cp}=1$ if the country is specialised in the activity p and 0 otherwise.

Relatedness (w_{cp}) is a number between 0 and 1, where 0 indicates that the product is not related to a country's current pattern of specialisation and 1 indicates that the product is highly related to the country's current pattern of specialisation.

3 Growth, Inequality & Emissions Models

Growth-Forecast Regression

To estimate the relationship between economic complexity and income growth, we use a panel regression model covering 174 countries for the periods 1999–2009 and 2009–2019. The dependent variable is the annualised growth rate of GDP per capita (in PPP, constant 2021 US dollars). The models control for the log of initial GDP per capita and include specifications with and without the Economic Complexity Index (ECI) and its interaction with initial income.

Table 1. Economic complexity and growth

Dependent variable:			
Annualised growth of GDP per capita (in PPP constant (USD) (1999-09, 2009-19)			
	(1)	(2)	(3)
ECI (trade)		0.778***	1.002***
		(0.197)	(0.232)
ECI (trade) x log of initial GDP per capita			-0.426**
			(0.197)
Log of initial GDP per capita	-0.714***	-1.289***	-1.421***
	(0.170)	(0.245)	(0.254)
Observations	174	174	174
R2	0.084	0.156	0.178
Adjusted R2	0.078	0.146	0.164
Residual Std Error	2.049 (df=172)	1.972 (df=171)	1.952 (df=170)
F Statistic	17.692*** (df=172)	13.830*** (df=171)	10.421*** (df=170)

Notes: Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

These coefficients are used to project the effect of changes in Rwanda's ECI on future GDP per capita growth, comparing a Core No-Gold baseline scenario with an ECI-OPT scenario. Projections use the estimated coefficients from column (3), applying Rwanda's observed or targeted ECI values and initial GDP per capita. All other variables are held at their observed values.

The regression results above provide the empirical foundation for these projections. For Rwanda, we set initial GDP per capita and ECI to their observed or target values to estimate expected growth rates under both the Core No-Gold baseline and the diversification scenarios.

Emissions-Forecast Regression

To quantify the relationship between economic complexity and environmental performance, we estimate panel regressions linking the Economic Complexity Index (ECI) to CO₂ emissions intensity (emissions per unit of GDP), using country-level data from 1995–2022. The dependent variable is the natural logarithm of emissions intensity, capturing how efficiently countries generate output relative to their carbon footprint.

Table 2. Economic complexity and emissions

Dependent variable:			
GHG emissions per unit of GDP, log (1996-99, 2000-03, 2004-07, 2008-11, 2012-15, 2016-18)			
ECI (trade)	(1)	(2)	(3)
		-0.240***	-0.236***
		(0.028)	(0.028)
ECI (trade), squared			-0.024***
			(0.017)
log of GDP per capita	0.199***	0.138***	0.248***
	(0.029)	(0.031)	(0.044)
Observations	489	489	489
R2	0.259	0.344	0.346
Adjusted R2	0.258	0.341	0.342
Residual Std. Error	0.492 (df=487)	0.464 (df=486)	0.463 (df=485)
F Statistic	164.559***	131.348***	90.106***

Notes: Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The coefficients from the preferred model (column 3) are used to estimate how Rwanda's planned increase in ECI would affect its emissions intensity.

The effect is:

- Rwanda's Core No-Gold baseline emissions intensity:
- Scenario:

A rise in Rwanda's ECI from -0.45 to +0.18 is expected to lower emissions intensity by approximately 21.6 per cent. This reduction reflects the benefits of structural transformation and complexity-driven industrial change, and it occurs even before considering shifts in energy sources.

Inequality-Forecast Regression

To estimate the relationship between economic complexity and income inequality, we use a panel regression model covering 320 country-periods from 1995–2022, with country fixed effects. The dependent variable is the Gini coefficient (net income). The models control for the log of initial GDP per capita and its square, and include specifications with ECI (trade) and its interaction with initial income.

Table 3. Economic complexity and inequality

Dependent variable:			
	GINI		
	(1)	(2)	(3)
ECI (trade)		–4.689***	–3.069***
		(0.527)	(0.491)
ECI (trade) x log of initial GDP per capita			–2.125***
			(0.494)
log of initial GDP per capita	0.199***	0.138***	–2.310***
	(0.029)	(0.031)	(0.523)
log of initial GDP per capita squared	0.087	0.170	1.154**
	(0.598)	(0.409)	(0.471)
Observations	320	320	320
R2	0.360	0.583	0.609
Adjusted R2	0.356	0.579	0.604
Residual Std. Error	5.023 (df=317)	4.061 (df=316)	3.938 (df=315)
F Statistic	67.861***	127.502***	98.951***

Notes: Standard errors in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

These coefficients are used to estimate the effect of changes in Rwanda's ECI on future income inequality, comparing a Core No-Gold baseline with a target ECI trajectory. Projections use the coefficients from column (3), applying Rwanda's observed or projected ECI and GDP per capita values. Other variables are held at their observed values.

For Rwanda, scenario calculations use the observed or targeted ECI and GDP per capita values, applying this equation to project expected Gini coefficients under different diversification scenarios.

4. Export-Portfolio Optimization (ECI-OPT)

To identify the most effective diversification strategy, this report applies the ECI-OPT method developed by Stojkoski and Hidalgo (2025). This approach moves beyond visual heuristics and directly solves for the optimal set of products that can raise Rwanda's Economic Complexity Index (ECI) while minimising the “effort” required to reach revealed comparative advantage (RCA) in new sectors.

Key Features of the Method

- **Target setting:** The method first defines a target ECI for Rwanda, based on empirical growth benchmarks and the current structure of exports.
- **Effort minimisation:** For each potential new product, the required “effort” is calculated as the incremental increase in RCA Rwanda would need to achieve by a set target year (e.g. from $RCA < 1$ to $RCA \geq 1$), using a stepping-stone year to model gradual progress.
- **Model inputs:** The forecast of future RCA uses a multivariate regression including (a) past RCA, (b) relatedness, (c) relative relatedness and (d) other control variables, all calibrated on historical trade data.
- **Optimisation process:** The algorithm searches for the smallest set of products which, if developed to $RCA \geq 1$, would raise the average complexity (PCI) of Rwanda's export basket above the target threshold. The cost function is the sum of the RCA increases required.
- **Forward-looking:** Unlike static diagrams, the model updates PCI values over time, adjusting for the possibility that some products may become less complex as more countries enter those activities.

Advantages Over Standard Approaches

- **Strategic alignment:** The method avoids sectoral biases and heuristics, focusing instead on the true cost-benefit trade-off for each product.
- **Balanced portfolio:** ECI-OPT naturally recommends a mix of related and unrelated activities, balancing feasibility and long-term complexity gains.
- **Empirical foundation:** Recommendations are grounded in observed patterns of trade evolution and robust out-of-sample forecasting performance.

Limitations

- The method assumes Rwanda is optimising in isolation, without fully accounting for general equilibrium effects if multiple countries pursue the same strategy.
- ECI is optimised as a target, so policymakers must also address the institutional and organisational capabilities needed to realise the recommended portfolio.

Application in This Report

- For Rwanda, the model identifies a set of 52 products that—if developed—would raise the ECI from -0.45 to $+0.18$ by 2033.
- Each product is scored for (a) effort required, (b) complexity gain and (c) expected contribution to ECI uplift.
- Products are then grouped into three strategic clusters to guide policy, investment and skills-development priorities.

Implementation Choices for Rwanda (project-specific)

Parameter	Value	Rationale
Target ECI	+0.18 (≈ ½ █ gain)	Matches a 5 % p.a. GDP-per-capita growth scenario
Base year tt	2023 trade matrix	Latest complete data
Stepping-stone █	5 years (2028)	Allows policy lead-time
Horizon ΔtΔt	10 years (2033)	
Complexity band	PCI≤1	92% of peer successes lie here
Logistics filter	Unit value higher than US\$1,654 per ton	Decreases the effects of high transportation costs
Objective	$\min \sum W_{c,p} \min \sum W_{c,p}$	Focus on least-cost upgrade

Outcome: 52 products raise ECI from –0.45 to 0.18.

Reference: Stojkoski, V., & Hidalgo, C.A. (2025). Optimizing Economic Complexity. Toulouse School of Economics Working Paper 1623.

5. Cluster Snapshot and Full Product List

To translate Rwanda’s diversification strategy into a concrete path for investment, the 52 products identified through the ECI-OPT model are grouped into three strategic clusters. Each cluster reflects a distinct development logic and is anchored by flagship products that exemplify the opportunity.

Cluster	What it means	Flagship products (HS-4)	Representative adjacencies	Core capabilities to (start/upgrade)
Foundational Upgrades	Quick wins from formulation/ blending, packaging, and quality control that unlock immediate market access.	3506 Glues & adhesives; 3819 Hydraulic brake fluids	3304 Beauty/make-up & skin-care preps; 3306 Oral / dental hygiene preps; 3003 Medicaments (bulk); 3808 Insecticides / repellents; 2939 Vegetable alkaloids	Batch formulation; in-process QC; packaging / labeling & traceability; basic GMP documentation; dimensional QC (packaging/plastics); line hygiene

Adjacent Growth	Step-ups that require reactors, dispersion / milling, plastics conversion, tooling, and certifications to enter higher-value supply chains.	3208 Paints & varnishes (aqueous); 3922 Plastic sanitary ware (baths / sinks / basins); 2922 Oxygen-function amino compounds	3917 Plastic tubes / pipes / hoses; 4010 Rubber conveyor / transmission belts; 7003 Cast/rolled glass; 6912 / 6914 Ceramic ware & other ceramic articles; 6815 Articles of mineral substances; 8303/8304 Safes & filing cabinets; 8311 Soldering / welding wire / rods; 4809 / 4816 / 4901 Coated / printing papers & books; 8452 Sewing machines; 7018 Glass beads / imitation stones	Reactor operations; dispersion/milling & film performance; tooling & mold maintenance; tolerancing/fixtures; ISO 9001/14001 supplier qualification; EHS compliance; packaging/print QA
Strategic Bets	Knowledge- and certification-intensive niches that anchor clean assembly, metrology, and regulated-market entry—pulling capabilities toward the dense core.	9018 Medical instruments; 8538 Parts for electrical apparatus; 8473 Office-machine / computer parts	3917 Medical-grade/coated tubing; sterile disposables & kits (adjacent to 9018, 3003); selected diagnostic components	Clean-room routines; sterilization & traceability (ISO 13485); documentation; precision finishing; regulated-market onboarding

Foundational Capability Upgrades (7)

- 2939 — Vegetable alkaloids
- 3003 — Medicaments (not in measured doses)
- 3304 — Beauty/make-up, skin-care preparations
- 3306 — Preparations for oral/dental hygiene
- 3506 — Prepared glues and other adhesives (Flagship)
- 3808 — Insecticides/rodenticides/fungicides/herbicides
- 3819 — Hydraulic brake fluids, anti-freezing preparations (Flagship)

Adjacent Growth Opportunities (34)

- 2823 — Titanium oxides
- 2922 — Oxygen-function amino-compounds (Flagship)
- 3208 — Paints & varnishes (aqueous medium) (Flagship)
- 3703 — Photographic paper/paperboard/textiles (coated media)
- 3917 — Plastic tubes, pipes, hoses
- 3922 — Plastic sanitary ware: baths/sinks/wash basins (Flagship)
- 4010 — Conveyor & transmission belts (vulcanised rubber)
- 4809 — Carbon paper, self-copy paper
- 4816 — Self-copy paper, duplicator stencils
- 4901 — Printed books, brochures
- 5401 — Sewing thread (man-made filaments)
- 5907 — Textile fabrics impregnated/covered for painting
- 5909 — Textile hosepiping & similar tubing
- 6601 — Umbrellas and sun-umbrellas
- 6812 — Fabricated asbestos fibers; mixtures with asbestos †
- 6815 — Articles of stone or other mineral substances
- 6912 — Ceramic table/kitchen-ware
- 6914 — Other ceramic articles
- 7003 — Cast/rolled glass
- 7018 — Glass beads; imitation stones
- 7418 — Copper table/kitchen articles
- 8203 — Files, rasps, pliers, scissors, hand tools
- 8211 — Knives with cutting blades
- 8303 — Safes/strong-boxes/cash boxes
- 8304 — Filing/card-index cabinets
- 8311 — Wire/rods/tubes/etc. of base metal for soldering/welding
- 8452 — Sewing machines
- 8474 — Machinery for sorting/screening/washing minerals
- 9207 — Keyboard musical instruments

- 9208 — Musical boxes/accordions
- 9608 — Ballpoint & felt-tip pens
- 9613 — Cigarette lighters
- 9617 — Vacuum flasks
- 9704 — Revenue stamps

Strategic Innovation Bets (11)

- 8473 — Office-machine/computer parts (Flagship)
- 8518 — Loudspeakers/headphones/microphones
- 8519 — Sound recording/reproducing equipment
- 8538 — Parts for electrical apparatus (8535–8537) (Flagship)
- 8539 — Electric filament/discharge lamps
- 8703 — Motor cars and other vehicles
- 8803 — Aircraft/spacecraft parts
- 9007 — Cinematographic cameras/projectors
- 9018 — Medical instruments (Flagship)
- 9025 — Hydrometers/thermometers/pyrometers/barometers
- 9402 — Medical/surgical/dental/veterinary furniture

Cluster	HS Code	Product name	Current RCA	Current Relative Relatedness	PCI	Target ECI	Target Growth
Adjacent Growth Opportunities	2823	Titanium oxides	0.203	-1.057	0.847	0.06	4.825
Adjacent Growth Opportunities	2922	Oxygen-function amino-compounds	0.3	-0.48	0.704	0.01	4.743
Foundational Capability Upgrades	2939	Vegetable alkaloids	0.844	-0.859	0.417	-0.34	4.17
Foundational Capability Upgrades	3003	Medicaments (not in measured doses)	0.287	0.186	0.645	0.16	4.989
Adjacent Growth Opportunities	3208	Paints & varnishes (aqueous medium)	0.458	0.732	0.587	-0.14	4.498
Foundational Capability Upgrades	3304	Beauty/make-up, skin-care preparations	0.366	0.104	0.515	0.17	5.006
Foundational Capability Upgrades	3306	Preparations for oral/dental hygiene	0.846	0.245	0.251	-0.39	4.088
Foundational Capability Upgrades	3506	Prepared glues and other adhesives	0.271	-0.44	0.853	-0.04	4.661
Adjacent Growth Opportunities	3703	Photographic paper, paperboard, textiles	0.151	-0.931	0.949	0.11	4.907
Foundational Capability Upgrades	3808	Insecticides, rodenticides, fungicides, herbicides	0.758	0.613	0.179	-0.29	4.252
Foundational Capability Upgrades	3819	Hydraulic brake fluids, anti-freezing preparations	0.422	-0.299	0.86	-0.19	4.416
Adjacent Growth Opportunities	3917	Plastic tubes, pipes, hoses	0.645	1.024	0.046	0.17	5.006
Adjacent Growth Opportunities	3922	Baths, sinks, wash basins, plastic sanitary ware	0.546	0.173	0.554	-0.14	4.498

Adjacent Growth Opportunities	4010	Conveyor belts, transmission belts, of vulcanised rubber	0.161	-0.327	0.938	0.06	4.825
Adjacent Growth Opportunities	4809	Carbon paper, self-copy paper	0.886	-0.488	0.394	-0.39	4.088
Adjacent Growth Opportunities	4816	Carbon paper, self-copy paper, duplicator stencils	0.204	-1.207	0.862	0.16	4.989
Adjacent Growth Opportunities	4901	Printed books, brochures	0.448	0.159	0.816	-0.19	4.416
Adjacent Growth Opportunities	5401	Sewing thread of man-made filaments	0.897	0.79	-0.322	-0.29	4.252
Adjacent Growth Opportunities	5907	Textile fabrics impregnated/covered with preparations for painting	0.651	-0.65	0.723	-0.29	4.252
Adjacent Growth Opportunities	5909	Textile hose piping and similar tubing	0.257	-0.699	0.892	-0.04	4.661
Adjacent Growth Opportunities	6601	Umbrellas and sun umbrellas	0.712	-0.491	0.043	-0.09	4.579
Adjacent Growth Opportunities	6812	Fabricated asbestos fibers; mixtures with asbestos	0.984	0.26	0.02	-0.39	4.088
Adjacent Growth Opportunities	6815	Articles of stone or other mineral substances	0.18	-0.483	0.96	0.01	4.743
Adjacent Growth Opportunities	6912	Ceramic tableware, kitchenware	0.448	-0.301	0.447	-0.09	4.579
Adjacent Growth Opportunities	6914	Ceramic articles	0.487	-0.256	0.391	0.06	4.825
Adjacent Growth Opportunities	7003	Cast glass and rolled glass	0.752	-1.238	0.849	-0.34	4.17

Adjacent Growth Opportunities	7018	Glass beads, imitation pearls, precious/semi-precious stones	0.212	-0.979	0.948	0.01	4.743
Adjacent Growth Opportunities	7418	Table, kitchen articles of copper	0.528	-0.639	0.345	0.01	4.743
Adjacent Growth Opportunities	8203	Files, rasps, pliers, scissors, hand tools	0.614	-0.678	0.685	-0.24	4.334
Adjacent Growth Opportunities	8211	Knives with cutting blades	0.277	-1.136	0.807	-0.04	4.661
Adjacent Growth Opportunities	8303	Safes, strong-boxes, cash/valuables boxes	0.216	-0.553	0.799	0.16	4.989
Adjacent Growth Opportunities	8304	Filing cabinets, card-index cabinets	0.403	-0.43	0.714	-0.09	4.579
Adjacent Growth Opportunities	8311	Wire, rods, tubes, etc. of base metal for soldering, welding	0.966	0.136	0.635	-0.44	4.006
Adjacent Growth Opportunities	8452	Sewing machines	0.902	-0.346	0.128	-0.39	4.088
Strategic Innovation Bets	8473	Parts and accessories for office machines, computers	0.147	-1.065	0.984	0.11	4.907
Adjacent Growth Opportunities	8474	Machinery for sorting, screening, washing, mineral substances	0.479	-0.11	0.645	-0.24	4.334
Strategic Innovation Bets	8518	Loudspeakers, headphones, microphones	0.611	-0.72	0.828	-0.29	4.252
Strategic Innovation Bets	8519	Sound recording/reproducing equipment	0.852	-0.85	0.824	-0.44	4.006
Strategic Innovation Bets	8538	Parts for electrical apparatus of 8535–8537	0.115	-0.255	0.997	0.06	4.825
Strategic Innovation Bets	8539	Electric filament/discharge lamps	0.352	-0.602	0.943	-0.09	4.579

Strategic Innovation Bets	8703	Motor cars and other vehicles	0.127	-0.381	0.951	0.11	4.907
Strategic Innovation Bets	8803	Parts of aircraft and spacecraft	0.71	-0.833	0.887	-0.34	4.17
Strategic Innovation Bets	9007	Cinematographic cameras and projectors	0.415	-0.002	0.813	-0.19	4.416
Strategic Innovation Bets	9018	Medical instruments	0.283	-0.52	0.892	-0.04	4.661
Strategic Innovation Bets	9025	Hydrometers, thermometers, pyrometers, barometers	0.142	-0.793	0.927	0.16	4.989
Adjacent Growth Opportunities	9207	Musical instruments, keyboards	0.599	-1.15	0.772	-0.24	4.334
Adjacent Growth Opportunities	9208	Musical boxes, fairground organs, accordions	0.922	-0.886	0.668	-0.44	4.006
Strategic Innovation Bets	9402	Medical, surgical, dental, veterinary furniture	0.143	-0.435	0.947	0.11	4.907
Adjacent Growth Opportunities	9608	Ballpoint pens, felt tipped pens	0.346	-0.747	0.691	-0.04	4.661
Adjacent Growth Opportunities	9613	Cigarette lighters	0.897	-0.271	0.075	-0.34	4.17
Adjacent Growth Opportunities	9617	Vacuum Flask	0.635	-1.225	0.472	-0.19	4.416
Adjacent Growth Opportunities	9704	Revenue Stamps	0.406	-0.211	0.841	-0.14	4.498

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All trade statistics (exports, imports) and calculations (RCA, PCI, Relatedness) use OEC data, February 2025 download.

